HANKEL DETERMINANTS FOR THE $(n+2)$-SHIFTED q-METALLIC NUMBERS

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ABSTRACT. Let $\Phi_n(q)$ denote the q-deformation (in the sense of Morier-Genoud & Ovsienko) of the metallic number $\phi_n = \frac{n+\sqrt{n^2+4}}{2}$. This is a power series with integer coefficients. We prove that the Hankel continued fraction of the $(n+2)$-shifted series $\Phi_n^{(n+2)}$ is eventually periodic, and that the corresponding Hankel determinant sequence $\Delta^{(n+2)}$ is $2n(n+1)$-periodic if n is even, $2n(n+1)$-antiperiodic if n is odd, and that it takes only the values $-2, -1, 0, 1, 2$. This confirms part of a former conjecture of ours and complements similar results we obtained for all ℓ-shifts of Φ_n with $0 \leqslant \ell \leqslant n+1$. In addition, we formulate a new conjecture of quasi-polynomiality for higher shifts $\ell \geqslant n+3$.

The determinant values arise as telescoping products of the partial numerators of the Hankel continued fraction. Most of the technically involved and lengthy verifications are carried out by a symbolic computation program, whose code is publicly available on GitHub for independent verification.

1. INTRODUCTION

The so-called *metallic numbers* (or *metallic ratios*, or *metallic means*) are the quadratic irrationals

$$\phi_n := \frac{n+\sqrt{n^2+4}}{2} = n + \frac{1}{n} + \frac{1}{n} + \frac{1}{n} + \cdots = n + \left(\frac{1}{n} + \right)^* \qquad (n \geqslant 1). \tag{1.1}$$

In particular, $\phi_1 = \frac{1+\sqrt{5}}{2}$ is the *golden ratio*. Here and thereafter, we use the following classical notation for infinite continued fractions:

$$\frac{\alpha_0}{\beta_0} + \frac{\alpha_1}{\beta_1} + \frac{\alpha_2}{\beta_2} + \cdots := \cfrac{\alpha_0}{\beta_0 + \cfrac{\alpha_1}{\beta_1 + \cfrac{\alpha_2}{\beta_2 + \cdots}}}$$

and indicate periodic data between parentheses and with a superscript $*$.

Thanks to S. Morier-Genoud and V. Ovsienko's fascinating theory of q-deformed real numbers (see [MGO20, MGO22] for the foundations, [MGO25] for a survey of its numerous occurrences in various branches of mathematics and [Ped26, Eti25, EO26] for recent developments on analytical

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aspects of q-numbers), one can attach to any metallic ratio ϕ_n an algebraic continued fraction $[\phi_n]_q$ called the *q-deformation* of ϕ_n.

By definition, $[\phi_n]_q$ is the following quantisation of the continued fraction (1.1):

$$[\phi_n]_q := [n]_q + \left(\frac{q^n}{[n]_{q^{-1}}} \; + \; \frac{q^{-n}}{[n]_q} \; + \right)^* \tag{1.2}$$

where the polynomial $[n]_q$ stands for the classical Euler-Gauss q-deformation of the integer n:

$$[n]_q := 1 + q + q^2 + \cdots + q^{n-1}.$$

As expected when speaking of quantisation, letting $q \to 1$ in (1.2) gives back expression (1.1) of ϕ_n. These algebraic continued fractions $[\phi_n]_q$ are called *the q-metallic numbers*.

A key fact of the q-deformation theory is that there exists a sequence of *integers* $(\kappa_l(\phi_n))_{l\geqslant 0}$, such that

$$[\phi_n]_q = \sum_{l=0}^{+\infty} \kappa_l(\phi_n) q^l,$$

i.e. $[\phi_n]_q$ expands about $q = 0$ as a formal power series in $\mathbb{Z}[[q]]$. Actually, this is a true power series expansion (with non-zero radius of convergence) when q is a complex variable, see [Eti25, EO26]. To view $[\phi_n]_q$ as a function of q we will use the following notation

$$\Phi_n(q) = [\phi_n]_q = \sum_{l=0}^{+\infty} \kappa_l(\phi_n) q^l \tag{1.3}$$

and still call the function $\Phi_n = \Phi_n(q)$ the n-th q-metallic number.

The q-metallic numbers are studied in depth in the articles [OP25], [HP25], [Ped26]. They reveal strong connections between the sequence of coefficients $\big(\kappa_l(\phi_n)\big)_{l\geqslant 0}$ in (1.3) and three highly famous sequences of the combinatorial world: the Catalan numbers, the Motzkin numbers and the enumeration of secondary structures of RNA molecules. These connections concern mainly the analytical properties of the sequences and their Hankel determinants.

About the latter, recall that for a power series $F(q) = \sum_{i=0}^{+\infty} f_i q^i$, or equivalently a sequence of numbers $F = (f_i)_{i\geqslant 0}$, the *Hankel determinants of F* are the doubly indexed family

$$\Delta_j^{(\ell)}(F) := \det\Big((f_{\ell+a+b})_{a,b=0}^{j-1} \Big) = \begin{vmatrix} f_\ell & f_{\ell+1} & \cdots & f_{\ell+j-1} \\ f_{\ell+1} & f_{\ell+2} & \cdots & f_{\ell+j} \\ \vdots & \vdots & & \vdots \\ f_{\ell+j-1} & f_{\ell+j} & \cdots & f_{\ell+2j-2} \end{vmatrix}, \qquad \ell, j \geqslant 0, \tag{1.4}$$

with convention

$$\Delta_0^{(\ell)}(F) := 1. \tag{1.5}$$

The determinant $\Delta_j^{(\ell)}(F)$ is of size $j \times j$, and we call the number ℓ the *shift* of the determinant, since it is clear that

$$\Delta_j^{(\ell)}\big((f_i)_{i\geqslant 0}\big) = \Delta_j^{(0)}\big((f_{i+\ell})_{i\geqslant 0}\big). \tag{1.6}$$

In other words,

$$\Delta_j^{(\ell)}(F) = \Delta_j^{(0)}(F^{(\ell)})$$

where $F^{(\ell)}(q)$ is the power series obtained by deleting the first ℓ Taylor coefficients of $F(q)$ and dividing by q^ℓ:

$$F^{(\ell)}(q) := \frac{1}{q^\ell}\left(F(q) - \sum_{i=0}^{\ell-1} f_i q^i\right) = \sum_{i=0}^{+\infty} f_{i+\ell} q^i. \tag{1.7}$$

When $\ell = 0$ the numbers $\Delta_j^{(0)}(F)$ will be called *ordinary Hankel determinants* of F. Often, we will write simply $\Delta^{(\ell)}(F)$ to denote the entire sequence $(\Delta_j^{(\ell)}(F))_{j\geqslant 0}$.

Also, we write, for short,

$$\Delta_j^{(\ell)} := \Delta_j^{(\ell)}(\Phi_n) = \Delta_j^{(0)}(\Phi_n^{(\ell)}).$$

It turns out that sequences $\Delta^{(\ell)}$, with $0 \leqslant \ell \leqslant n+1$, enjoy many striking properties, among which we emphasize the following (see [OP25] for case $n = 1, 2$ and [HP25] for general $n \geqslant 2$):

(I) Periodicity. The sequences $\Delta^{(\ell)}$, with $0 \leqslant \ell \leqslant n+1$, are $2n(n+1)$-periodic when n is even and $2n(n+1)$-antiperiodic when n is odd.

(II) Values. They take the values $-1, 0, 1$ only. The distribution of these values in $\Delta^{(0)}$ is governed by the symmetry

$$\Delta_j = (-1)^{n(n+1)/2}\,\Delta_{(2n+1)(n+1)-j} \qquad \text{for } 0 \leqslant j \leqslant (2n+1)(n+1),$$

and the shifted sequences are obtained from $\Delta^{(0)}$ by the contiguity relation

$$\Delta_j^{(\ell+1)} = (-1)^{j+\frac{n(n+2\ell-1)}{2}}\Delta_{j+n+1}^{(\ell)} \quad \text{for all } j \geqslant 0.$$

(III) Recurrence. The sequences $\Delta^{(\ell)}$, $0 \leqslant \ell \leqslant n+1$, all satisfy the same *Somos–Gale–Robinson recurrence*:

$$\Delta_{j+2n+2}^{(\ell)}\,\Delta_j^{(\ell)} = \Delta_{j+2n+1}^{(\ell)}\,\Delta_{j+1}^{(\ell)} - \left(\Delta_{j+n+1}^{(\ell)}\right)^2 \quad \text{for all } j \geqslant 0. \tag{1.8}$$

Hence they form periodic discrete integrable dynamical systems[1].

All these results are based on the expansion of the functions $\Phi_n^{(\ell)}$ (for $0 \leqslant \ell \leqslant n+1$) as a special type of continued fraction introduced in [Han16], called *Hankel continued fraction*, whose partial numerators determine all the Hankel determinants of the series; see Section 3 for the precise definition.

The main purpose of this paper is to treat the case $\ell = n+2$, technically much more difficult than for the lower shifts $0 \leqslant \ell \leqslant n+1$, and to make a precise conjecture as concerns the shifts $\ell \geqslant n+3$, which behave in a completely different way (see below).

Concerning case $\ell = n+2$, we will mainly prove the analogues of properties (I) & (II) above by following the same path, whose first step is:

Theorem A. *Let $n \geqslant 1$ be an integer. The Hankel continued fraction of the shifted q-metallic number $\Phi_n^{(n+2)}$ is eventually periodic:*

$$\Phi_n^{(n+2)}(q) = \mathcal{I}_n(q) \;{}_{+}\; \Big(\mathcal{P}_n(q)\Big)^*.$$

[1]We refer to [HP25] for some information on Somos-Gale-Robinson recurrences, which will not be developed here.

More precisely, the pre-periodic block $\mathcal{I}_n(q)$ consists of one single term, given by

$$\mathcal{I}_1(q) = \frac{-1}{1+2q}, \qquad \mathcal{I}_n(q) = \frac{q^{n-2}}{(q^n - Q)/(q-1)} \quad (n \geqslant 2),$$

with

$$Q := q^2 - q + 1,$$

and the periodic block $\mathcal{P}_n(q)$ consists of p_n terms, where

$$p_n := \begin{cases} 2 & \text{if } n = 1, \\ 9 & \text{if } n = 2, \\ 9n - 13 & \text{if } n \geqslant 3 \text{ is odd}, \\ 9n - 10 & \text{if } n \geqslant 4 \text{ is even}. \end{cases}$$

Note that the period p_n need not be the minimal one (although it is always the case in our numerical examples).

The complete Hankel continued fraction of $\Phi_n^{(n+2)}$ produced by Theorem A being too complicated to be written in the general case, let us provide a few examples.

Case $n = 1$, $p_n = 2$:

$$\Phi_1^{(3)}(q) = \frac{-1}{1+2q} \; + \; \left(\frac{-q^4}{1+q-q^2+2q^3} \; + \; \frac{-q^4}{1+2q} \right)^*$$

Case $n = 2$, $p_n = 9$:

$$\Phi_2^{(4)}(q) = \frac{1}{1} \; + \; \left(\frac{2q^2}{1+\frac{1}{2}q} \; + \; \frac{\frac{1}{4}q^2}{1+\frac{3}{2}q} \; + \; \frac{4q^2}{1-\frac{3}{2}q} \; + \; \frac{-\frac{1}{4}q^2}{1-\frac{3}{2}q} \right.$$
$$\left. + \; \frac{4q^2}{1+\frac{3}{2}q} \; + \; \frac{\frac{1}{4}q^2}{1+\frac{1}{2}q} \; + \; \frac{2q^2}{1} \; + \; \frac{q^5}{1+2q^2-q^3} \; + \; \frac{q^5}{1} \right)^*.$$

Case $n = 3$, $p_n = 14$:

$$\Phi_3^{(5)}(q) = \frac{q}{1+q^2} \; + \; \left(\frac{2q^3}{1+\frac{1}{2}q} \; + \; \frac{\frac{1}{4}q^2}{1-\frac{1}{2}q} \; + \; \frac{2q^2}{1} \; + \; \frac{-q^2}{1+q} \; + \; \frac{-q^3}{1-q+2q^2} \right.$$
$$+ \; \frac{-q^5}{1+q+q^2+2q^3} \; + \; \frac{-q^5}{1-q+2q^2} \; + \; \frac{-q^3}{1+q} \; + \; \frac{-q^2}{1}$$
$$\left. + \; \frac{2q^2}{1-\frac{1}{2}q} \; + \; \frac{\frac{1}{4}q^2}{1+\frac{1}{2}q} \; + \; \frac{2q^3}{1+q^2} \; + \; \frac{q^7}{1+q^2+2q^3-q^4} \; + \; \frac{q^7}{1+q^2} \right)^*.$$

Case $n=4$, $p_n=26$:

$$\begin{aligned}\Phi_4^{(6)}(q) = \frac{q^2}{1+q^2+q^3} \;+\; \Bigg(&\frac{2q^4}{1+\frac{1}{2}q} \;+\; \frac{\frac{1}{4}q^2}{1-\frac{1}{2}q} \;+\; \frac{2q^3}{1+q^2} \;+\; \frac{-q^3}{1+q} \;+\; \frac{-q^3}{1-q+q^2} \\ &+\; \frac{q^3}{1} \;+\; \frac{q^2}{1} \;+\; \frac{-q^2}{1+q} \;+\; \frac{2q^2}{1-\frac{1}{2}q} \;+\; \frac{-\frac{1}{4}q^2}{1+\frac{3}{2}q} \;+\; \frac{4q^2}{1-2q} \;+\; \frac{\frac{1}{2}q^3}{1+q+q^2} \\ &+\; \frac{\frac{1}{2}q^3}{1-2q} \;+\; \frac{4q^2}{1+\frac{3}{2}q} \;+\; \frac{-\frac{1}{4}q^2}{1-\frac{1}{2}q} \;+\; \frac{2q^2}{1+q} \;+\; \frac{-q^2}{1} \;+\; \frac{q^2}{1} \\ &+\; \frac{q^3}{1-q+q^2} \;+\; \frac{-q^3}{1+q} \;+\; \frac{-q^3}{1+q^2} \;+\; \frac{2q^3}{1-\frac{1}{2}q} \;+\; \frac{\frac{1}{4}q^2}{1+\frac{1}{2}q} \\ &+\; \frac{2q^4}{1+q^2+q^3} \;+\; \frac{q^9}{1+q^2+q^3+2q^4-q^5} \;+\; \frac{q^9}{1+q^2+q^3} \Bigg)^*.\end{aligned}$$

Our second main result concerns the $(n+2)$-shifted Hankel sequence of the q-metallic number $\Phi_n(q)$, i.e., the ordinary Hankel sequence of the $(n+2)$-shifted q-metallic number $\Phi_n^{(n+2)}(q)$.

Theorem B. *Let $n \geqslant 1$ be an integer. The sequence $\Delta^{(n+2)} = \big(\Delta_j^{(n+2)}\big)_{j\geqslant 0}$ of $(n+2)$-shifted Hankel determinants of the q-metallic number $\Phi_n(q)$ satisfies the following properties:*

1) Periodicity. *The sequence is $2n(n+1)$-periodic when n is even and $2n(n+1)$-antiperiodic (hence $4n(n+1)$-periodic) when n is odd;*

2) Values. *It consists of the values $-2,-1,0,1,2$ only (except if $n=1$: it can only take the values $-1,0,1$);*

Thus, the (anti)periodicity property is exactly the same as for the lower shifts $0 \leqslant \ell \leqslant n+1$, see (I) above. On the contrary the set of values is strictly bigger (except for $n=1$) but still remarkable. For instance, one period of $\Delta^{(n+2)}$ for $n=2$, and one antiperiod for $n=3$, read respectively

$$1,1,-2,-1,2,-1,-2,1,1,0,0,0,$$
$$1,0,-1,2,1,-1,1,0,-1,0,0,-1,0,1,-1,1,2,-1,0,1,0,0,0,0.$$

Let us make two comments on Theorem B. Firstly, it proves part (1) of [HP25, Conjecture E]; the case $n=1$ was already proved in [OP25, Theorem 1.6], but we treat here all cases $n \geqslant 1$. Secondly, we mention that we have sought for three-term Somos–Gale–Robinson recurrences of type (1.8)

$$\begin{aligned}&\Delta_j\Delta_{j+k} = \alpha\Delta_{j+a}\Delta_{j+k-a} + \beta\Delta_{j+b}\Delta_{j+k-b},\\ &\text{with}\quad 1 \leqslant a < b \leqslant \frac{k}{2},\quad \alpha\beta \neq 0\end{aligned}$$

which could be satisfied by the sequence $\Delta^{(n+2)}$, but we could not find any for $n = 1, \dots, 5$ and $k \leqslant 40$. Nor did we find any four-term recurrence

$$\Delta_j \Delta_{j+k} = \alpha\, \Delta_{j+a}\Delta_{j+k-a} + \beta\, \Delta_{j+b}\Delta_{j+k-b} + \gamma\, \Delta_{j+c}\Delta_{j+k-c},$$
$$\text{with} \quad 1 \leqslant a < b < c \leqslant \frac{k}{2}, \quad \alpha\beta\gamma \neq 0,$$

for $n = 2, \dots, 5$ and $6 \leqslant k \leqslant 40$.

A Hankel determinant sequence which is at the same time periodic and confined to the fixed finite set $\{-1, 0, 1\}$ (or even $\{-2, -1, 0, 1, 2\}$) is a rather unusual phenomenon. We found only two examples in the literature: [SX08] and [Bar12, Example 6]. In our case, the situation is all the more remarkable because a single further shift $\ell = n + 3$ of the sequence already completely changes everything. Indeed, we conjectured in [HP25] that all sequences $\Delta^{(\ell)}$ were unbounded if $\ell \geqslant n+3$. Computer experimentation now leads us to formulate a much more precise conjecture:

Conjecture C. Let $n \geqslant 1$ be an integer. For any $\ell \geqslant n + 3$, the shifted Hankel determinants $\Delta_j^{(\ell)} = \Delta_j^{(\ell)}(\Phi_n)$ are quasi-polynomials of period

$$\rho_n = \begin{cases} 2n(n+1) & \text{if } n \text{ is even,} \\ 4n(n+1) & \text{if } n \text{ is odd.} \end{cases}$$

More precisely, there exists a family $\big(Q_{\ell,r}\big)_{r=0}^{\rho_n - 1}$ of polynomials in $\mathbb{Q}[X]$, taking integer values on the integers, such that for all $0 \leqslant r \leqslant \rho_n - 1$ and $m \geqslant 0$,

$$\Delta^{(\ell)}_{r+m\rho_n} = Q_{\ell,r}(m).$$

Note finally that, *modulo a prime number* the Hankel determinants sequences $\Delta^{(\ell)}(\Phi_n)$ are periodic for *every* $n \geqslant 1$ and *every shift* $\ell \geqslant 0$: see [HP25, Theorem F].

Strategy and organisation of the paper.

To establish Theorem A we use the approach developed in [Han16]. The main tool is an algorithm `NextABC` which builds the Hankel continued fraction (H-fraction, for short) of a quadratic power series one step at a time. This is explained in Section 2, together with the quadratic equation verified by the shifted power series $\Phi_n^{(n+2)}$ of the q-deformed metallic ratio (Proposition 2.7).

Section 3 is devoted to the construction of the H-fraction of $\Phi_n^{(n+2)}$ and to the proof of Theorem A; this is the most technical part of the paper. At each step m, the algorithm `NextABC` produces a quintuple $(A_m, B_m, C_m; K_m, D_m)$ of polynomials, from which the m-th partial numerator and denominator of the H-fraction are read off. It turns out that the quintuple at index $p_n + 2$ coincides with the quintuple at index 2, and this gives the periodicity. But the law which governs the quintuples is complicated, and not even uniform in m: for $n \geqslant 4$, the quintuples belong either to two families, called *passes*, made of repeating blocks of nine formulas (one for each residue of m modulo 9), or to three short lists of quintuples, called *bridges*, which occur only once per period. These explicit lists, the *pass-tables*, are given in the appendices, and we prove that they are correct. Because of the size of the expressions, this proof uses SageMath programs, which certify the formulas symbolically, uniformly in n and m. The small cases $n = 1, 2, 3$ are simpler and are treated separately.

In Section 4 we prove Theorem B by applying [Han16, Theorem 2.1] (recalled as Theorem 2.2 below), which expresses the Hankel determinants of a power series in terms of the partial numerators of its H-fraction. These partial numerators are known from the pass-tables, but some care is needed to show that all the nonzero determinants belong to $\{-2,-1,1,2\}$ and to determine the sign of the (anti)periodicity.

Finally, Appendices A–G list the explicit tables for the 2 passes and the 3 bridges that are used in Section 3 to prove the periodicity of the Hankel continued fraction of $\Phi_n^{(n+2)}$.

Note. All the SageMath programs, their output logs, and the pass-table data are available at

`https://github.com/GuoniuHan/metallicplus`

We emphasise that these programs deliver rigorous symbolic proofs of our general statements, valid for all n and j at once, rather than checks of finitely many cases.

2. Background material and setup

2.1. **The Hankel continued fraction and the algorithm `NextABC`.** We first recall from [Han16] the definition and the basic property of Hankel continued fractions. Throughout, the base field is $\mathbb{Q}$.

Definition 2.1. For each positive integer δ, a *super continued fraction associated with* δ, called *super δ-fraction* for short, is a continued fraction of the form

$$F(q) = \frac{v_0\, q^{\sigma_0}}{1+qu_1(q)} + \frac{-v_1\, q^{\sigma_0+\sigma_1+\delta}}{1+qu_2(q)} + \frac{-v_2\, q^{\sigma_1+\sigma_2+\delta}}{1+qu_3(q)} + \frac{-v_3\, q^{\sigma_2+\sigma_3+\delta}}{1+qu_4(q)} + \cdots \tag{2.1}$$

where the $v_j \neq 0$ are constants, the σ_j are nonnegative integers, and the $u_j(q)$ are polynomials of degree less than or equal to $\sigma_{j-1}+\delta-2$ (by convention, 0 has degree -1).

When $\delta=1$ (resp. $\delta=2$) and all $\sigma_j=0$, the super δ-fraction (2.1) is the classical S-fraction (resp. J-fraction). A super 2-fraction is called a *Hankel continued fraction*, or *H-fraction* for short.

The following result will be the main tool used in this article.

Theorem 2.2 ([Han16, Theorem 2.1])**.** *1) Let δ be a positive integer. Each super δ-fraction defines a power series, and conversely, for each power series $F(q)$ the super δ-fraction expansion of $F(q)$ exists and is unique.*

2) Let $F(q)$ be a power series whose H-fraction is given by (2.1) *with $\delta=2$. Then the nonvanishing Hankel determinants of $F(q)$ are exactly those indexed by $s_j := \sigma_0+\sigma_1+\cdots+\sigma_{j-1}+j$ for every $j\geqslant 0$, and their values are given by the formula*

$$\Delta_{s_j}\big(F(q)\big) = (-1)^{\epsilon_j}\, v_0^{s_j} v_1^{s_j-s_1} v_2^{s_j-s_2}\cdots v_{j-1}^{s_j-s_{j-1}}, \tag{2.2}$$

where $\epsilon_j = \sum_{i=0}^{j-1}\sigma_i(\sigma_i+1)/2$.

Notice that the notion of H-fraction is also connected to *Artin's regular continued fraction*, see [HP25, §2.2].

Because we are only interested in H-fractions, from now on we fix $\delta = 2$ and set the following synthetic notation:

$$\mathcal{H}\big(q;(\sigma_j),(v_j),(u_j)\big) := \frac{v_0\,q^{\sigma_0}}{1+qu_1(q)} \;+\; \frac{-v_1\,q^{\sigma_0+\sigma_1+2}}{1+qu_2(q)} \;+\; \frac{-v_2\,q^{\sigma_1+\sigma_2+2}}{1+qu_3(q)} \;+\; \cdots \tag{2.3}$$

We recall from [Han16, Algorithm 3.1] (see also [HP25, §3.1]) how the H-fraction can be constructed algorithmically for power series $F(q) \in \mathbb{Q}[[q]]$ satisfying a quadratic equation

$$A(q) + B(q)F(q) + C(q)F(q)^2 = 0 \tag{2.4}$$

with $A, B, C \in \mathbb{Q}[q]$ such that

$$A \neq 0, \quad B(0) = 1, \quad C \neq 0, \quad C(0) = 0. \tag{2.5}$$

Algorithm 2.3. [Algorithm `NextABC` for H-fractions]

`Prototype:` $(A^*, B^*, C^*; k, a, D) = \texttt{NextABC}(A, B, C)$.

`Input:` $A(q), B(q), C(q) \in \mathbb{Q}[q]$ three polynomials satisfying conditions (2.5).

`Output:` an integer $k \geqslant 0$,
a nonzero scalar $a \in \mathbb{Q}$,
a polynomial $D \in \mathbb{Q}[q]$ of degree $\leqslant k+1$ such that $D(0) = 1$,
three polynomials $A^*, B^*, C^* \in \mathbb{Q}[q]$ satisfying (2.5).

`Step 1:` [Define k, a]. Since $A(q) \neq 0$, let $A(q) = aq^k + O(q^{k+1})$ with $a \neq 0$.

`Step 2:` [Define D]. Define $D(q)$ by

$$\frac{aq^kB}{A} - \frac{aq^kC}{B} = D(q) + O(q^{k+2}), \tag{2.6}$$

where $D(q)$ is a polynomial of degree at most $k+1$; then $D(0) = 1$.

`Step 3:` [Define A^*, B^*, C^*]. Let

$$A^*(q) = \big(-D^2A/a + BD\,q^k - Ca\,q^{2k}\big)/q^{2k+2}, \tag{2.7}$$

$$B^*(q) = 2AD/(a\,q^k) - B, \tag{2.8}$$

$$C^*(q) = -A\,q^2/a. \tag{2.9}$$

These are polynomials in $\mathbb{Q}[q]$ satisfying (2.5) by [Han16, Lemma 3.2].

Remark 2.4. In Step 2, if $C = q^2C'$ then $aq^kB/A = D(q) + O(q^{k+2})$; if moreover $A = aq^k$, then $B = D(q) + O(q^{k+2})$.

Let us explain how and why algorithm `NextABC` computes the H-fraction of F.

• We initialise the data by setting $F_1 := F$ and $(A_1, B_1, C_1) := (A, B, C)$. From (2.4), $F_1 = -A_1/(B_1 + C_1F_1)$, so by (2.5) and Step 1,

$$F_1(q) = -a\,q^k + O(q^{k+1}),$$

where aq^k is the lowest term of A_1. Hence $-aq^k/F_1 = 1+O(q)$, and there are a unique polynomial D of degree at most $k+1$ and a unique power series F_2 such that

$$\frac{-a\,q^k}{F_1} = D(q) - q^{k+2}F_2(q), \qquad \text{i.e.} \qquad F_1 = \frac{-a\,q^k}{D(q) - q^{k+2}F_2(q)}. \tag{2.10}$$

This D is the polynomial of Step 2: indeed

$$\frac{-a\,q^k}{F_1} = \frac{a\,q^k B_1}{A_1} + \frac{a\,q^k C_1 F_1}{A_1}$$

with

$$\frac{a\,q^k C_1 F_1}{A_1} = \frac{-a\,q^k C_1}{B_1 + C_1 F_1} = \frac{-a\,q^k C_1}{B_1} + O(q^{2k+2}),$$

since $F_1 = O(q^k)$ and $C_1 = O(q)$.

• Now, apply Step 3 of `NextABC`: define A^*, B^*, C^* by (2.7), (2.8), (2.9), respectively. According to [Han16, Lemma 3.2], for these polynomial conditions (2.5) still hold and the power series F_2 is still quadratic, namely:

$$A^* + B^* F_2 + C^* F_2^2 = 0.$$

Thus the algorithm can be applied to F_2, and so on.

• Writing $(A_{m+1}, B_{m+1}, C_{m+1})$ for the triple of F_{m+1}, we thus define a sequence

$$\big(A_{m+1}, B_{m+1}, C_{m+1};\, k_m, a_m, D_m\big) = \texttt{NextABC}\big(A_m, B_m, C_m\big), \qquad m \geqslant 1,$$

which gives the iterated version of (2.10): $F_m = -a_m q^{k_m} / \big(D_m - q^{k_m+2} F_{m+1}\big)$. Eventually, we obtain the H-fraction of F:

$$\begin{aligned} F(q) &= \frac{-a_1 q^{k_1}}{D_1} + \frac{a_2\, q^{k_1+k_2+2}}{D_2} + \frac{a_3\, q^{k_2+k_3+2}}{D_3} + \cdots \\ &= \mathcal{H}\big(q; (\sigma_j), (v_j), (u_j)\big) \end{aligned} \tag{2.11}$$

where we use notation (2.3) with $v_j = -a_{j+1}$ and $\sigma_j = k_{j+1} = \operatorname{val} A_{j+1}$ for $j \geqslant 0$, and $1 + u_j(q) q = D_j$ for $j \geqslant 1$.

To shorten the notation we will record k_m and a_m simultaneously in the leading monomial

$$K_m := a_m q^{k_m}$$

of A_m, so that Step 1 reads $A_m = K_m + O(q^{k_m+1})$. In other words, the H-fraction (2.11) reads also

$$F(q) = \frac{-K_1(q)}{D_1(q)} + \frac{K_2(q)\, q^{k_1+2}}{D_2(q)} + \frac{K_3(q)\, q^{k_2+2}}{D_3(q)} + \cdots \tag{2.12}$$

Definition 2.5. The sequence of polynomial quintuples

$$x_m := \big(A_m, B_m, C_m; K_m, D_m\big), \qquad m \geqslant 1,$$

produced by Algorithm `NextABC` from the quadratic equation (2.4) and conditions (2.5), is called the *sequence of quintuples associated with* F. By (2.11) it determines the H-fraction of F, and therefore, by Theorem 2.2(2), all its Hankel determinants. Note that all the entries of x_m carry the same index m, whereas Algorithm `NextABC` applied to (A_m, B_m, C_m) returns

$$(A_{m+1}, B_{m+1}, C_{m+1}; k_m, a_m, D_m).$$

Note that x_{m+1} depends only on (A_m, B_m, C_m). Consequently, if $(A_{p'}, B_{p'}, C_{p'}) = (A_p, B_p, C_p)$ for some indices $p < p'$, then the sequence of quintuples is periodic from index p on, with period $p' - p$, and the H-fraction of F is eventually periodic. This is precisely what is going to happen for $F = \Phi_n^{(n+2)}$, with $p = 2$ and $p' = p_n + 2$, where p_n is given in Theorem A; see Section 3.

2.2. **The q-metallic number and its shifted series.** Let us recall some facts about the q-metallic power series Φ_n (see [OP25, HP25, Ped26]). Almost by definition (1.2) and (1.3), the q-metallic power series Φ_n is characterised by the quadratic functional equation

$$q\,\Phi_n(q)^2 + \big((1+q^n)(1-q) - q[n]_q\big)\Phi_n(q) = 1. \tag{2.13}$$

from which follows at once an explicit formula:

$$\Phi_n(q) = \frac{1}{2q}\left(q[n]_q + (q^n+1)(q-1) + \sqrt{\big(q[n]_q + (q^n+1)(q-1)\big)^2 + 4q}\right).$$

Although we will not use it, we mention that the Taylor series expansion about zero of Φ_n is of the form:

$$\Phi_n(q) = 1 + q + \cdots + q^{n-1} + q^{2n} + \sum_{i=2n+1}^{+\infty} \kappa_i(\phi_n) q^i,$$

where coefficients $\kappa_i(\phi_n)$ are integers; see Corollary 2.6 in [OP25].

Recall that we are interested in the ℓ-shifted Hankel determinants of Φ_n (for $\ell \geqslant 0$), or equivalently in the ordinary Hankel determinants of the shifts $\Phi_n^{(\ell)}$ (definition (1.7)):

$$\Delta_j^{(\ell)} := \Delta_j^{(\ell)}(\Phi_n) = \Delta_j\big(\Phi_n^{(\ell)}\big).$$

Specifically, our target is the shift $\ell = n+2$, i.e. the series $\Phi_n^{(n+2)}$.

The starting point of our whole method is that each shift is *again* a quadratic power series, obtained from the previous one by an elementary substitution. More precisely, one has the following statement, easy to check and already proved in [HP25, Lemma 7.1].

Lemma 2.6. *Let $F(q)$ be a power series with constant term f_0, satisfying a quadratic equation*

$$A + BF + CF^2 = 0$$

with A, B, C polynomials, $C \neq 0$. Then the shift $F^{(1)}(q) = \dfrac{F(q) - f_0}{q}$ satisfies the quadratic equation

$$A' + B'F^{(1)} + C'\big(F^{(1)}\big)^2 = 0,$$

where

$$A' = \frac{1}{q}\big(A + f_0 B + f_0^2 C\big), \qquad B' = B + 2f_0 C, \qquad C' = qC. \tag{2.14}$$

From now on we put:

$$Q := q^2 - q + 1.$$

Iterating Lemma 2.6 from (2.13) produces the quadratic equation of every shift $\Phi_n^{(\ell)}$. Indeed, rewriting

$$A^{(0)} + B^{(0)}\Phi_n + C^{(0)}\Phi_n^2 = 0$$

for (2.13) in the normalisation

$$A^{(0)} = -1, \quad B^{(0)} = (1+q^n)(1-q) - q[n]_q, \quad C^{(0)} = q,$$

one proves by induction (see [HP25, Prop. 7.2]) that

$$A^{(\ell)} + B^{(\ell)}\Phi_n^{(\ell)} + C^{(\ell)}\big(\Phi_n^{(\ell)}\big)^2 = 0,$$

where, for $0 \leqslant \ell \leqslant n$,

$$A^{(\ell)} = \frac{q^{\ell+1} - Q\,(q^n - q^{n-\ell} + 1)}{(q-1)^2}, \quad B^{(\ell)} = \frac{2q^{\ell+1} - Q\,(q^n+1)}{q-1}, \quad C^{(\ell)} = q^{\ell+1}, \tag{2.15}$$

while at $\ell = n+1$

$$A^{(n+1)} = -q^{n-1}, \qquad B^{(n+1)} = -\frac{Q + (q^2 - 3q + 1)q^n}{q-1}, \qquad C^{(n+1)} = q^{n+2}. \tag{2.16}$$

The case we need, $\ell = n+2$, is one further step of Lemma 2.6.

Proposition 2.7. *For every integer $n \geqslant 1$, the shifted series $F = \Phi_n^{(n+2)}$ is a quadratic power series with integer coefficients, satisfying*

$$A^{(n+2)} + B^{(n+2)}\,F + C^{(n+2)}\,F^2 = 0, \tag{2.17}$$

where

$$\begin{cases} A^{(n+2)} = Q, \quad B^{(n+2)} = 2q^3 - q^2 + q + 1, \quad C^{(n+2)} = q^4, & \text{if } n = 1, \\ A^{(n+2)} = -q^{n-2}, \quad B^{(n+2)} = -\dfrac{(q^2 - 3q + 1)\,q^n + Q}{q-1}, \quad C^{(n+2)} = q^{n+3} & \text{if } n \geqslant 2. \end{cases} \tag{2.18}$$

Moreover $B^{(n+2)}, C^{(n+2)} \in \mathbb{Z}[q]$ and the triple $(A^{(n+2)}, B^{(n+2)}, C^{(n+2)})$ satisfies the conditions (2.5).

Proof. We start with case $n \geqslant 2$ and apply Lemma 2.6 to coefficients (2.16) of the series $F = \Phi_n^{(n+1)}$. The constant term $f_0 = \Phi_n^{(n+1)}(0)$ is determined by evaluating $A^{(n+1)} + B^{(n+1)} f_0 + C^{(n+1)} f_0^2 = 0$ at $q = 0$: since $n \geqslant 2$ we have $A^{(n+1)}(0) = 0$, $C^{(n+1)}(0) = 0$ and $B^{(n+1)}(0) = 1$, whence $f_0 = 0$. Thus (2.14) degenerates to

$$A^{(n+2)} = \frac{1}{q}\,A^{(n+1)} = -q^{n-2}, \quad B^{(n+2)} = B^{(n+1)}, \quad C^{(n+2)} = q\,C^{(n+1)} = q^{n+3},$$

which is the second line of (2.18). The numerator of $B^{(n+2)}$ vanishes at $q = 1$, so $B^{(n+2)} \in \mathbb{Z}[q]$. For $n = 1$ the same computation runs one step longer. Formula (2.16) still holds, giving

$$A^{(2)} = -1, \quad B^{(2)} = 1 + q - q^2, \quad C^{(2)} = q^3,$$

but now the constant term of $\Phi_1^{(2)}$ is $f_0 = 1$, and Lemma 2.6 yields the first line of (2.18). In both cases one checks easily $A^{(n+2)} \neq 0$, $C^{(n+2)} \neq 0$, $B^{(n+2)}(0) = 1$ and $C^{(n+2)}(0) = 0$. □

3. The Hankel continued fraction of $\Phi_n^{(n+2)}$: proof of Theorem A

This section is devoted to the proof of Theorem A. Following the method explained in Section 2.1, we will show that the sequence of quintuples :

$$x_m = (A_m, B_m, C_m; K_m, D_m), \qquad m = 1, 2, 3, \ldots, \tag{3.1}$$

associated with $F := \Phi_n^{(n+2)}$ (in the sense of Definition 2.5) and starting from the triple

$$(A_1, B_1, C_1) := \big(A^{(n+2)}, B^{(n+2)}, C^{(n+2)}\big)$$

of Proposition 2.7, is such that

$$x_{j+kp_n} = x_j \quad \text{for all } j \geqslant 2 \text{ and } k \geqslant 0, \tag{3.2}$$

where p_n is the period defined by

$$p_n := \begin{cases} 2 & \text{if } n = 1, \\ 9 & \text{if } n = 2, \\ 9n - 13 & \text{if } n \geqslant 3 \text{ is odd}, \\ 9n - 10 & \text{if } n \geqslant 4 \text{ is even}. \end{cases} \tag{3.3}$$

This will imply all assertions of Theorem A: the first quintuple x_1 determines the pre-periodic block $\mathcal{I}_n(q)$ in the H-fraction of $\Phi_n^{(n+2)}$, while the set of quintuples x_m for $2 \leqslant m \leqslant p_n + 1$ determines the periodic block $\mathcal{P}_n(q)$.

This section is organised as follows: §3.1 gives a criterion, valid for any quadratic series, for checking one step of Algorithm `NextABC`; §3.2 treats $n \leqslant 3$; §3.3 describes the tables of the appendices and defines from them a sequence (y_m) for $n \geqslant 4$; §3.4 proves that $(y_m) = (x_m)$; and §3.5 concludes.

3.1. **Verification of one step of Algorithm `NextABC`.** For a triple (A, B, C) and a quintuple $x = (A, B, C; K, D)$ with $K = a\,q^k$, $a \in \mathbb{Q}^*$, $k \in \mathbb{Z}$, $D \in \mathbb{Q}[q]$, we shall use the conditions

$$A, B, C \in \mathbb{Q}[q], \quad B(0) = 1, \quad C(0) = 0, \tag{3.4}$$

$$k \geqslant 0, \quad D(0) = 1, \quad \deg D \leqslant k + 1. \tag{3.5}$$

Condition (3.4) is the input condition (2.5) of Algorithm `NextABC` apart from $A, C \neq 0$, and every output triple (A^*, B^*, C^*) of Algorithm `NextABC` satisfies (3.4).

For two quintuples $x = (A, B, C; K, D)$, $K = aq^k$, and $x' = (A', B', C'; K', D')$ we write

$$x \to x' \qquad \text{if} \qquad \texttt{NextABC}(A, B, C) = (A', B', C';\, k, a, D),$$

that is, if (k, a, D) is the output of Steps 1 and 2 of Algorithm `NextABC` applied to (A, B, C), and (A', B', C') is the triple then computed by Step 3. By construction, the sequence of quintuples associated with F is the unique sequence (x_m) such that x_1 has the triple (A_1, B_1, C_1) and $x_m \to x_{m+1}$ for all $m \geqslant 1$.

Step 3 (2.7)–(2.9) makes sense for any quintuple, whereas Steps 1 and 2 are truncations, which depend on the sizes of the exponents. The following criterion replaces them by a polynomiality condition.

Lemma 3.1. *Let $x = (A, B, C; K, D)$, $K = aq^k$, satisfy* (3.4) *and* (3.5)*, with $C \neq 0$, and define A^*, B^*, C^* by* (2.7)–(2.9)*. Then*

$$\texttt{NextABC}(A, B, C) = (A^*, B^*, C^*;\, k, a, D)$$

if and only if $A^*, B^* \in \mathbb{Q}[q]$ and $B^*(0) = 1$.

Proof. The direct implication was noted above. Conversely, assume that $A^*, B^* \in \mathbb{Q}[q]$ and $B^*(0) = 1$.

Step 1. By (2.8), $2AD = aq^k(B + B^*)$, where $B + B^*$ is a polynomial with constant term $1 + 1 = 2$. Since $D(0) = 1$, it follows that $A = aq^k + O(q^{k+1})$. So Step 1 of Algorithm `NextABC`,

applied to A, gives exactly k and a. (In particular $A \neq 0$; indeed, if $A = 0$ then $B^* = -B$ and $B^*(0) = -1$.)

Step 2. Multiplying (2.7) by $-a/A$ gives

$$D^2 - \frac{aq^k B}{A}\, D + \frac{a^2 q^{2k} C}{A} = -\frac{aq^k}{A}\, q^{k+2} A^*.$$

Since $aq^k/A = 1 + O(q)$ is a power series and A^* is a polynomial, the right-hand side is $O(q^{k+2})$; so D is a solution of the congruence

$$X^2 - \frac{aq^k B}{A}\, X + \frac{a^2 q^{2k} C}{A} \equiv 0 \pmod{q^{k+2}}. \tag{3.6}$$

Let $\widehat{D}$ be the polynomial defined by Step 2, i.e. $\widehat{D} \equiv aq^k B/A - aq^k C/B \pmod{q^{k+2}}$. Since $aq^k B/A \equiv \widehat{D} + aq^k C/B$, the left-hand side of (3.6) at $X = \widehat{D}$ is congruent modulo q^{k+2} to

$$\widehat{D}^2 - \Big(\widehat{D} + \frac{aq^k C}{B}\Big)\widehat{D} + \frac{a^2 q^{2k} C}{A} = aq^k C\Big(\frac{aq^k}{A} - \frac{\widehat{D}}{B}\Big).$$

Here $aq^k C = O(q^{k+1})$ because $C(0) = 0$, and the bracket is $O(q)$ because aq^k/A, $\widehat{D}$ and B all have constant term 1. Hence $\widehat{D}$ is also a solution of (3.6). Subtracting the two congruences gives

$$(D - \widehat{D})\Big(D + \widehat{D} - \frac{aq^k B}{A}\Big) \equiv 0 \pmod{q^{k+2}},$$

and the second factor has constant term $1 + 1 - 1 = 1$, so it is invertible in $\mathbb{Q}[[q]]$. Therefore $D \equiv \widehat{D} \pmod{q^{k+2}}$, and $D = \widehat{D}$ because both have degree at most $k+1$. Thus (k, a, D) is the output of Steps 1 and 2, and Step 3 then produces (A^*, B^*, C^*). □

Since $C^* = -Aq^2/a$ is a polynomial with $C^*(0) = 0$ as soon as A is a polynomial, the lemma gives the following practical test.

Corollary 3.2. *Let $x = (A, B, C; K, D)$ and $x' = (A', B', C'; K', D')$ be two quintuples, with $C \neq 0$. Assume that*

(S3) *the triple computed from x by Step 3 is (A', B', C');*

(S12) *(A, B, C) and (A', B', C') satisfy* (3.4)*, and x satisfies* (3.5)*.*

Then $x \to x'$.

Here, condition (S3) (meaning 'Step 3') is an identity between rational functions, whereas (S12) (meaning 'Steps 1 & 2') is a set of statements about q-orders.

Conversely, if $x \to x'$ then (A', B', C') satisfies (3.4). Hence, if x satisfies (3.5) and the triple computed from x by Step 3 does *not* satisfy (3.4), then the K, D of x are not those of Algorithm `NextABC`.

3.2. **The cases $n = 1, 2, 3$.** For a fixed numerical n, Algorithm `NextABC` can simply be run from the initial triple $(A_1, B_1, C_1) = \big(A^{(n+2)}, B^{(n+2)}, C^{(n+2)}\big)$ of Proposition 2.7. For $n = 1, 2, 3$ this is done by the program `verify_base.sage`, and the resulting quintuples are listed in Appendices A, B and C. In each case one observes that

$$(A_{p_n+2}, B_{p_n+2}, C_{p_n+2}) = (A_2, B_2, C_2)$$

with $p_n = 2, 9, 14$ respectively, so that the sequence is periodic from $m = 2$ on, with period p_n; moreover $x_{1+p_n} = (-A_1, B_1, -C_1; *, *)$.

3.3. **The pass-tables for $n \geqslant 4$.** From now on $n \geqslant 4$. In this subsection we only *describe* the tables of Appendices D–H and use them to define a sequence of quintuples $(y_m)_{m\geqslant 1}$; that $y_m = x_m$ is proved in §3.4. Let us first introduce some notation and vocabulary.

- The entries of the tables are rational functions of q, q^{2j} and q^n. An index m is written $m = 9j + r$ with $r \in \{1, \dots, 9\}$: the *block* $j \geqslant 0$ consists of the indices $9j+1, \dots, 9j+9$, and r is the position in the block. We put
$$d := n - 2j.$$
- A *pass* is a family of nine formulas, the *rows* $r = 1, \dots, 9$, giving the quintuple at index $9j+r$ as a function of (n, j). There are two passes:
  - *pass* 1 (Appendix D), whose rows are given separately for $j = 0$ and $r \leqslant 6$ (the *special rows*) and by a *generic row* otherwise;
  - *pass* 2 (Appendix E), whose rows are all generic; its index $9j + r$ is local to the pass, and j is an integer when n is even, a half-integer when n is odd (in both cases $d = n - 2j$ is even).
- There are also three *bridges*, which are short lists of quintuples free of j: we call them $\mathrm{b}1, \dots, \mathrm{b}8$ (Appendix F), $\mathrm{c}1, \dots, \mathrm{c}5$ (Appendix G) and $\mathrm{d}1, \mathrm{d}2, \mathrm{d}3$ (Appendix H). All the formulas hold for every $n \geqslant 4$, even or odd.
- A *tuple* is the value of a row, or a bridge entry, at a given (n, j).
- Inside a pass, the *successor* of row $r < 9$ of block j is row $r + 1$ of block j, and the successor of row 9 of block j is row 1 of block $j + 1$.

Now, let us describe the sequences (y_m) in the pass-tables of the Appendices.

- *The sequence (y_m) for even $n \geqslant 4$.* Put $t_n := \frac{9}{2}n - 7$ and recall that $p_n = 9n - 10$.
  (a) For $1 \leqslant m \leqslant t_n - 1$, y_m is row r of pass 1 at block j, where $m = 9j + r$. Thus pass 1 runs through the full blocks $j = 0, \dots, \frac{n}{2} - 2$ (that is, $d = n, n-2, \dots, 4$), and through row 1 of the block $j = \frac{n}{2} - 1$ ($d = 2$).
  (b) $y_{t_n}, \dots, y_{t_n+4}$ are $\mathrm{c}1, \dots, \mathrm{c}5$, respectively.
  (c) The next $\frac{9}{2}n - 14$ terms are the rows of pass 2 from local index 6 (row 6 at $j = 0$) to local index $\frac{9}{2}n - 9$ (row 9 at $j = \frac{n}{2} - 2$, i.e. $d = 4$), in increasing order.
  (d) The next eight terms are $\mathrm{b}1, \dots, \mathrm{b}8$; thus $y_{1+p_n} = \mathrm{b}8$.
  (e) $y_{m+p_n} = y_m$ for all $m \geqslant 2$.
- *The sequence (y_m) for odd $n \geqslant 5$.* Put $t_n := \frac{9n-15}{2}$ and recall that $p_n = 9n - 13$.
  (a) For $1 \leqslant m \leqslant t_n - 1$, y_m is row r of pass 1 at block j, where $m = 9j + r$. Thus pass 1 runs through the full blocks $j = 0, \dots, \frac{n-5}{2}$ ($d = n, n-2, \dots, 5$), and through rows 1–5 of the block $j = \frac{n-3}{2}$ ($d = 3$).
  (b) $y_{t_n}, y_{t_n+1}, y_{t_n+2}$ are $\mathrm{d}1, \mathrm{d}2, \mathrm{d}3$, respectively.
  (c) The next $\frac{9n-29}{2}$ terms are the rows of pass 2, read at half-integer j, from local index $\frac{13}{2}$ (row 2 at $j = \frac{1}{2}$) to local index $\frac{9}{2}n - 9$ (row 9 at $j = \frac{n}{2} - 2$, i.e. $d = 4$), in increasing order.

(d) The next eight terms are b1, ..., b8; thus $y_{1+p_n} = \mathrm{b8}$.
(e) $y_{m+p_n} = y_m$ for all $m \geqslant 2$.

In both cases the counts add up: one period ($m = 2, \dots, 1 + p_n$) consists of $t_n - 2$ tuples of pass 1, the tuples of the bridge c (resp. d), the tuples of pass 2 and the eight tuples of the bridge b, i.e.

$$\left(\tfrac{9}{2}n - 9\right) + 5 + \left(\tfrac{9}{2}n - 14\right) + 8 = 9n - 10 = p_n$$

for even n and

$$\tfrac{9n-19}{2} + 3 + \tfrac{9n-29}{2} + 8 = 9n - 13 = p_n$$

for odd n. For a later use, we observe also that

$$y_1 = \left(-q^{n-2},\, B_1,\, q^{n+3};\, -q^{n-2},\, D\right), \qquad \mathrm{b8} = \left(q^{n-2},\, B_1,\, -q^{n+3};\, q^{n-2},\, D\right), \tag{3.7}$$

with $D = -\frac{Q-q^n}{q-1}$. Hence y_1 has the triple (A_1, B_1, C_1) of Proposition 2.7, and b8 has the triple $(-A_1, B_1, -C_1)$.

3.4. **Verification of the tables.**

Proposition 3.3. *Let $n \geqslant 4$. For every $m \geqslant 1$, $y_m \to y_{m+1}$. Hence $y_m = x_m$ for all $m \geqslant 1$.*

Proof. By condition (e) above (for both parities), it suffices to treat $1 \leqslant m \leqslant 1 + p_n$. We apply Corollary 3.2 to each pair (y_m, y_{m+1}), and check its two conditions.

Condition (S3). The pairs (y_m, y_{m+1}) are of finitely many kinds: a row of a pass and its successor (generic rows, special rows, or the step from the last special row to the generic rows), and the steps involving a bridge, namely

$$\begin{array}{llll} \text{pass 1, row 1 at } d = 2 \to \mathrm{c1}, & \mathrm{c5} \to \text{pass 2, row 6 at } j = 0 & (n \text{ even}), \\ \text{pass 1, row 5 at } d = 3 \to \mathrm{d1}, & \mathrm{d3} \to \text{pass 2, row 2 at } j = \tfrac{1}{2} & (n \text{ odd}), \\ \text{pass 2, row 9 at } d = 4 \to \mathrm{b1}, & \mathrm{b}i \to \mathrm{b}(i{+}1), \qquad \mathrm{b8} \to y_2. & \end{array}$$

- For a step from a row to its successor, (S3) is an identity between rational functions of q, $u := q^{2j}$ and $w := q^n$; since it does not depend on the relative size of the exponents, it is an identity in $\mathbb{Q}(q, u, w)$, valid for all n and j at once (note $q^{4j} = u^2$ and $q^{n-2j} = w/u$).
- For a step into a bridge, (S3) says that the triple computed by Step 3 from the last row of the pass is that of the bridge entry. By the identity of the first case, Step 3 maps this row to the triple of the next row, so (S3) amounts to the identity saying that the bridge entry has the triple of the next row of the pass at the last block, e.g. that c1 has the triple of row 2 of pass 1 at $q^{2j} = q^{n-2}$.
- For a step out of a bridge (c5, d3, bi), (S3) is a direct identity, the target row being taken at a fixed j.

Thus (S3) is an identity in $\mathbb{Q}(q, u, w)$ in the first case and in $\mathbb{Q}(q, w)$ in the other two; all these identities are proved by the program `certify_step3.sage` by reading the tables from `passtables.sage`.

Condition (S12). This is a set of statements about q-orders. Write each entry X of a tuple as a finite sum

$$X = \sum_{b,c} P_{b,c}(q)\, u^b w^c, \qquad u = q^{2j},\ w = q^n,\ P_{b,c} \in \mathbb{Q}(q).$$

In the tables, (b,c) ranges over $(0,0),(1,0),(-1,1),(0,1)$ — the terms $1, q^{2j}, q^{n-2j}, q^n$ — and the denominator of $P_{b,c}$ has the form $\lambda\, q^s (q-1)^e$ with $\lambda \in \mathbb{Q}^*$ and s, e two nonnegative integers. The term (b,c) has q-order

$$\operatorname{ord}(b,c) = \operatorname{val}_q P_{b,c} + 2jb + nc, \tag{3.8}$$

an affine function of (j,n). An entry X is a polynomial in q as soon as (a) the factor $(q-1)^e$ of its denominator divides the numerator after the substitution $u = q^{2j}$, $w = q^n$, and (b) every term has $\operatorname{ord}(b,c) \geqslant 0$. Condition (a) is checked identically in j and n, by expanding the numerator at $q = 1+\varepsilon$ with j, n symbolic (the coefficients of $\varepsilon^0, \dots, \varepsilon^{e-1}$ are polynomials in j, n, and they vanish). Condition (b), and likewise $B(0) = 1$ (B starts with the term 1 and every other term has order $\geqslant 1$), $C(0) = 0$, $k \geqslant 0$, $D(0) = 1$ and $\deg D \leqslant k+1$, are inequalities between affine functions of (j,n), to be proved on a set of the form $\mathcal{C} = \{\, j \geqslant j_0,\ n - 2j \geqslant c \,\}$.

But, for an affine function $g = \alpha + \beta j + \gamma n$ and a target $\tau \in \mathbb{Z}$, substituting $j = j_0 + s$ and $n - 2j = c + s'$ with $s, s' \geqslant 0$ gives the finite test

$$g(j,n) \geqslant \tau \ \text{ on } \mathcal{C} \iff \beta + 2\gamma \geqslant 0, \quad \gamma \geqslant 0, \quad \alpha + \beta j_0 + \gamma(2j_0 + c) \geqslant \tau. \tag{3.9}$$

The program `certify_steps12.sage` proves in this way, for every tuple of the tables, that the tuple satisfies (3.5) and that its triple, as well as the triple computed from it by Step 3, satisfy (3.4), in the following ranges: every special row and every bridge entry, for all $n \geqslant 4$; row r of pass 1 (resp. pass 2), at every (n,j) with $d \geqslant c_r$ (resp. $d \geqslant c'_r$) at which it occurs in the definition of (y_m), where

$$(c_1, \dots, c_9) = (1,3,3,2,0,4,4,3,1), \qquad (c'_1, \dots, c'_9) = (3,3,2,0,2,4,4,3,1).$$

No value of n or j is ever substituted.

Coverage. It remains to check that every tuple is used in (y_m) only inside its range. All the bounds c_r, c'_r are at most 4, and the full blocks used in (a) and (c) have $d \geqslant 4$. The incomplete last block of pass 1 uses row 1 at $d = 2$ (even n; $c_1 = 1$) or rows 1–5 at $d = 3$ (odd n; $c_1, \dots, c_5 \leqslant 3$). The j-free tuples are used only for $n \geqslant 4$. Hence Corollary 3.2 applies to every pair (y_m, y_{m+1}).

Finally y_1 has the triple (A_1, B_1, C_1), by (3.7). Since $y_m \to y_{m+1}$ for every m, induction on m gives $y_m = x_m$. □

Worked instance. Take $m = 9j + 3$ in pass 1 with $j \geqslant 1$. The table (Appendix D) reads

$$A = 2q^{d-3}, \qquad K = 2q^{d-3}, \qquad B = -\frac{Q}{q-1}\,(1+q^n) - \frac{2(q-2)}{q(q-1)}\,q^d,$$

$$C = \frac{Qq^2(q-2)}{2(q-1)^2}\,(1+q^n) + \frac{q(q-2)^2}{2(q-1)^2}\,q^d + \frac{Qq^4}{2(q-1)^2}\,q^{2j}, \qquad D = \frac{q^{d-1}-Q}{q-1} = [d-1]_q - q.$$

The orders of the terms of A are $d-3$; of B, 0 (with $-Q/(q-1) = 1 + O(q)$), n and $d-1$; of C, 2, $n+2$, $d+1$ and $2j+4$. With the cancellation of $(q-1)^e$, the triple satisfies (3.4) for $d \geqslant 3$.

Next, $k = d-3 \geqslant 0$, $D(0) = 1$ and $\deg D \leqslant d-2 = k+1$ (with equality for $d \geqslant 4$; for $d = 3$, $D = 1$). Step 3 gives

$$A^* = -Q\,[n]_q - q^{d-1} + Q\,q^{2j+1}, \qquad B^* = Q\,[n]_q + 2q^{d-1},$$

which are the A, B of row 4; they are polynomials, and $B^*(0) = 1$ for $d \geqslant 2$. Hence the row $\to$ row 4 on $\{j \geqslant 1,\ d \geqslant 3\}$, i.e. $c_3 = 3$.

Remark 3.4 (Why a pass ends)**.** The bridges are forced by Corollary 3.2. For instance, for even n, at the last block $d = 2$ of pass 1, row 1 $\to$ the quintuple with the triple of row 2; but the triple of row 3 does not satisfy (3.4) there ($A = 2q^{d-3} = 2q^{-1}$), so the K, D of row 2 at $d = 2$ are not those of Algorithm `NextABC` (end of §3.1), and the quintuple at index t_n must be given by another formula: this is c1, which has the same triple as row 2 at $d = 2$ but its own K, D. Similarly, for odd n, row 7 of pass 1 has $A = q^{d-4}$, not a polynomial at $d = 3$; and in pass 2, row 2 has $A = Q\,q^{d-3}$, not a polynomial at $d = 2$. The header $n - 2j \geqslant h_r$ of row r in the appendices gives the exact range on which the triple of this generic row satisfies (3.4), except for row 1 of each pass, whose triple satisfies (3.4) also at $n - 2j = -1$ (a value that never occurs).

3.5. **Proof of Theorem A.**

Proposition 3.5. *For every $n \geqslant 1$, the sequence of quintuples (x_m) associated with $\Phi_n^{(n+2)}$ is periodic from $m = 2$ on, with period p_n (defined in Theorem A), and x_{1+p_n} has the triple $(-A_1, B_1, -C_1)$.*

Proof. For $n \leqslant 3$ see §3.2. For $n \geqslant 4$, $x_m = y_m$ by Proposition 3.3, and (y_m) is periodic from $m = 2$ with period p_n by construction; $x_{1+p_n} =$ b8 has the triple $(-A_1, B_1, -C_1)$ by (3.7). □

Notice that the pre-period is exactly 1: if x_1 recurred, it would recur at index $1 + p_n$, but $x_{1+p_n} \neq x_1$.

Proof of Theorem A. By (2.12), the H-fraction of $F = \Phi_n^{(n+2)}$ is

$$F = \frac{\alpha_0}{\beta_0} \;+\; \frac{\alpha_1}{\beta_1} \;+\; \frac{\alpha_2}{\beta_2} \;+\; \cdots,$$

with $\alpha_0 = -K_1$, $\beta_0 = D_1$, and $\alpha_i = K_{i+1}\,q^{k_i+2}$, $\beta_i = D_{i+1}$ for $i \geqslant 1$. For $i \geqslant 1$, the term α_i/β_i depends only on x_i and x_{i+1}. By Proposition 3.5, $x_{i+p_n} = x_i$ for $i \geqslant 2$, and $k_{1+p_n} = \operatorname{val} A_{1+p_n} = \operatorname{val} A_1 = k_1$; hence $\alpha_{i+p_n}/\beta_{i+p_n} = \alpha_i/\beta_i$ for all $i \geqslant 1$. This is the claimed form, with $\mathcal{P}_n(q)$ the block of the terms $i = 1, \ldots, p_n$, and $\mathcal{I}_n(q) = -K_1/D_1$.
Moreover, for $n \geqslant 2$, (3.7) gives $K_1 = -q^{n-2}$ and $D_1 = (q^n - Q)/(q-1)$; for $n = 1$, Appendix A gives $K_1 = 1$ and $D_1 = 1 + 2q$. □

4. The $(n+2)$-shifted Hankel determinants: proof of Theorem B

4.1. **Proof of Theorem B.2: the values.** Theorem A reduces part 2 of Theorem B to a finite computation with the data (σ_j, v_j) over one period: these are given by finitely many affine formulas in n and j, and we must show that the resulting Hankel determinants, computed with (2.2), lie in $\{-2, -1, 0, 1, 2\}$.

The starting point is Theorem 2.2 (part 2) that we apply to the H-fraction (2.11) of $F = \Phi_n^{(n+2)}$ just built in previous section. With the data (σ_j, v_j) of (2.11), formula (2.2) says that the only nonzero Hankel determinants of F are the $\Delta_{s_j}(F)$ at the indices

$$s_0 = 0, \qquad s_j = \sigma_0 + \sigma_1 + \cdots + \sigma_{j-1} + j \quad (j \geqslant 1), \tag{4.1}$$

and there

$$\Delta_{s_j}(F) = (-1)^{\epsilon_j}\, v_0^{s_j}\, v_1^{s_j - s_1} \cdots v_{j-1}^{s_j - s_{j-1}}, \qquad \epsilon_j = \sum_{i=0}^{j-1} \binom{\sigma_i + 1}{2}. \tag{4.2}$$

All other Hankel determinants of F vanish; these account for the value 0 in Theorem B. It therefore remains to show that each product (4.2) equals ± 1 or ± 2.

Recall that $v_i = -a_{i+1}$ and $\sigma_i = k_{i+1}$. Thus the pair (v_i, σ_i) is read off (in the pass-tables in the Appendices) from the leading monomial $K_{i+1} = a_{i+1} q^{k_{i+1}}$ of the quintuple x_{i+1}, and it is clear that the partial numerators v_i of the H-fraction are all of the form

$$v_i = \pm 2^{e_i}, \quad \text{with } e_i \in \mathbb{Z}.$$

Therefore the product (4.2) is

$$\Delta_{s_j}(F) = (-1)^{\epsilon_j} \Big(\prod_{i<j} (\operatorname{sgn} v_i)^{\, s_j - s_i} \Big)\, 2^{E_j}, \qquad \text{where} \quad E_j := \sum_{i=0}^{j-1} e_i\, (s_j - s_i). \tag{4.3}$$

Thus $\Delta_{s_j}(F) \in \{\pm 2, \pm 1\}$ if and only if $E_j \in \{0, 1\}$ for every j (the sign is automatically $\pm$). In other words, Theorem B.2 is equivalent to the purely arithmetic statement

$$E_j \in \{0, 1\} \qquad \text{for all } j \geqslant 0. \tag{4.4}$$

Recall $s_{i+1} - s_i = \sigma_i + 1$, and let

$$T_i := \sum_{l=0}^{i} e_l$$

be the partial sums of the exponent sequence. Reversing the order of summation in (4.3),

$$E_j = \sum_{i=0}^{j-1} e_i \sum_{l=i}^{j-1} (\sigma_l + 1) = \sum_{l=0}^{j-1} (\sigma_l + 1) \sum_{i=0}^{l} e_i = \sum_{l=0}^{j-1} (\sigma_l + 1)\, T_l. \tag{4.5}$$

Lemma 4.1. *For every $n \geqslant 1$ the data (σ_i, e_i) of the H-fraction of $\Phi_n^{(n+2)}$ satisfy:*

(i) *$T_i \in \{-1, 0, 1\}$ for all $i \geqslant 0$;*

(ii) *$\sigma_i = 0$ whenever $T_i \neq 0$;*

(iii) *the partial sums $\sum_{l=0}^{j-1} T_l$ lie in $\{0, 1\}$ for all $j \geqslant 0$.*

Proof. Since T changes only at the indices with $e_i \neq 0$, and $\sum_{l<j} T_l$ changes only at the indices with $T_l \neq 0$, it suffices to list the strings of consecutive indices at which $e_i \neq 0$, and to check that before each string $T = 0$ and $\sum T = 0$, and that along it properties (i), (ii) and (iii) hold and both sums return to 0. In all cases only the following three strings of exponents occur; we

give for each the values of T and of $\sum_{l\leqslant i} T_l$ along the string (starting from 0), and the letters at which $T \neq 0$, where (ii) requires $\sigma = 0$:

(e_i)	(T_i)	$\left(\sum_{l\leqslant i} T_l\right)$	$\sigma = 0$ at letters
$1, -2, 1$	$1, -1, 0$	$1, 0, 0$	$1, 2$
$1, -2, 2, -1, -1, 2, -2, 1$	$1, -1, 1, 0, -1, 1, -1, 0$	$1, 0, 1, 1, 0, 1, 0, 0$	$1, 2, 3, 5, 6, 7$
$1, -2, 2, -2, 2, -2, 1$	$1, -1, 1, -1, 1, -1, 0$	$1, 0, 1, 0, 1, 0, 0$	$1, \ldots, 6$

(4.6)

Between two strings all e_i vanish, so T and $\sum T$ stay at 0.

For $n = 1$ all partial numerators are ± 1 (Appendix A), so $e_i \equiv 0$ and there is nothing to prove. For $n = 2$, by Appendix B, $e_0 = 0$ and the exponents over one period ($i = 1, \ldots, 9$) are $1, -2, 2, -2, 2, -2, 1, 0, 0$, the third string of (4.6), with $\sigma_1 = \cdots = \sigma_6 = 0$. For $n = 3$, by Appendix C, $e_0 = 0$ and over one period ($i = 1, \ldots, 14$) the nonzero exponents are two copies of the first string, at $i = 1, 2, 3$ and $i = 10, 11, 12$, with $\sigma_1 = \sigma_2 = \sigma_{10} = \sigma_{11} = 0$.

Let $n \geqslant 4$. We follow the sequence $(x_m) = (y_m)$ of §3.3, and read a_m (hence e_{m-1}, defined by $a_m = \pm 2^{e_{m-1}}$) and k_m ($= \sigma_{m-1}$) from the values of $K_m = a_m q^{k_m}$ in the tables. In every tuple, a is a constant, independent of n and j.

Generic rows. For the generic rows of a pass, a depends only on the residue r (for pass 2, r is the residue in the index $9j + r$ of Appendix E), and k is affine in j and $d = n - 2j$:

r	1	2	3	4	5	6	7	8	9
a, pass 1	2	$-\frac{1}{4}$	2	-1	1	1	1	1	-1
k, pass 1	0	0	$d-3$	0	$2j-1$	1	$d-4$	0	$2j$
e, pass 1	1	-2	1	0	0	0	0	0	0
a, pass 2	1	1	1	-1	2	$-\frac{1}{4}$	2	-1	1
k, pass 2	1	$d-3$	0	$2j-1$	0	0	$d-4$	0	$2j$
e, pass 2	0	0	0	0	1	-2	1	0	0

So in a full block of either pass the nonzero exponents form the first string of (4.6) (rows $1, 2, 3$ of pass 1, rows $5, 6, 7$ of pass 2), with $k = 0$ at its first two letters.

The other tuples. The rows $m = 1, \ldots, 6$ of pass 1 at $j = 0$ and the bridges have

	m=1	2	3	4	5	6	c1	c2	c3	c4	c5	d1	d2	d3
a	-1	2	$\frac{1}{4}$	2	-1	-1	$-\frac{1}{4}$	4	$\frac{1}{2}$	$\frac{1}{2}$	4	1	-1	-1
k	$n-2$	0	0	$n-3$	0	1	0	0	$n-3$	0	0	1	$n-1$	1
e	0	1	-2	1	0	0	-2	2	-1	-1	2	0	0	0

	b1	b2	b3	b4	b5	b6	b7	b8
a	1	-1	-1	2	$\frac{1}{4}$	2	1	1
k	1	0	$n-3$	0	0	$n-2$	$n+1$	$n-2$
e	0	0	0	1	-2	1	0	0

Along the sequence. At $m = 1$, $e_0 = 0$. The rows $m = 2, 3, 4$ give the first string, with $k = 0$ at $m = 2, 3$, and the subsequent tuples of pass 1 give zeros and copies of the first string, as above. For odd n this goes on up to row 5 of the last block ($d = 3$); then d1, d2, d3 and the rows

$2, 3, 4$ of pass 2 at $j = \frac{1}{2}$ give zeros. For even n, the last block of pass 1 ($d = 2$) contributes only row 1, and row 1 at $d = 2$, c1, ..., c5 and the rows 6, 7 of pass 2 at $j = 0$ give the exponents $1, -2, 2, -1, -1, 2, -2, 1$, the second string of (4.6), with $k = 0, 0, 0, n-3, 0, 0, 0, n-4$, hence $k = 0$ at letters $1, 2, 3, 5, 6, 7$. In both cases pass 2 then gives zeros and copies of the first string, up to row 9 of its last full block. Finally b1, ..., b8 give $0, 0, 0, 1, -2, 1, 0, 0$, the first string with $k = 0$ at b4, b5, and the sequence returns to $m = 2$. This covers one period, and the lemma follows. □

Theorem B.2 is an immediate consequence of this lemma. Indeed, by (ii), $(\sigma_l + 1)T_l = T_l$ for every l, so (4.5) collapses to

$$E_j = \sum_{l=0}^{j-1} T_l, \tag{4.7}$$

which lies in $\{0, 1\}$ by (iii). Thus (4.4) holds.

4.2. **Proof of Theorem B.1: the (anti-)periodicity.** In this section we explain how the p_n-periodicity of the H-fraction of $\Phi_n^{(n+2)}$ — or, equivalently, of the associated sequence of quintuples $x_m = (A_m, B_m, C_m; K_m, D_m)$ — yields the $2n(n+1)$-periodicity (if n is even) or $2n(n+1)$-antiperiodicity (if n is odd) of the Hankel determinant sequence $\Delta^{(n+2)}$.

As usual, we let $n \geqslant 1$ be any integer and, to make the text easier to read, we put

$$L := 2n(n+1) \quad \text{and} \quad P := p_n.$$

By Proposition 3.5, $x_{m+P} = x_m$ for $m \geqslant 2$, and x_{1+P} has the triple $(-A_1, B_1, -C_1)$. Changing the signs of A and C does not change k, changes a into $-a$, and does not change D in Step 2 of Algorithm `NextABC`. Hence $K_{1+P} = -K_1$, and in terms of the H-fraction this means:

$$\sigma_{i+P} = \sigma_i \quad (i \geqslant 0), \qquad v_{i+P} = v_i \quad (i \geqslant 1), \qquad v_P = -v_0. \tag{4.8}$$

In particular,

$$s_{i+P} = s_i + s_P, \qquad \epsilon_{i+P} = \epsilon_i + \epsilon_P, \qquad e_{i+P} = e_i \quad \text{for all } i \geqslant 0. \tag{4.9}$$

Lemma 4.2. *One has: $s_P = L$ and $s_{i+P} = s_i + L$ for all $i \geqslant 0$.*

Proof. Because of (4.9) it suffices to prove first assertion. By definition,

$$s_P = \sum_{m=0}^{P-1} (\sigma_m + 1) = \sum_{m=1}^{P} (k_m + 1).$$

For $n \geqslant 4$, summing the values of k in the tables segment by segment (one full block of either pass contributes $9 + (2n-7) = 2n+2$, by the k-table in the proof of Lemma 4.1) gives

segment	n even	n odd
$m = 1, \dots, 6$ $(j = 0)$	$2n+2$	$2n+2$
pass 1, rows $7, 8, 9$ at $j = 0$	$n-1$	$n-1$
full blocks of pass 1	$(\frac{n}{2} - 2)(2n+2)$	$\frac{n-5}{2}(2n+2)$
last block of pass 1	1 (row 1)	$n+1$ (rows 1–5)
bridge c, resp. d	$n+2$	$n+4$
pass 2 before its full blocks	n (rows 6–9, $j = 0$)	$2n$ (rows 2–9, $j = \frac{1}{2}$)
full blocks of pass 2	$(\frac{n}{2} - 2)(2n+2)$	$\frac{n-5}{2}(2n+2)$
b1, …, b7	$3n+4$	$3n+4$
total	$2n(n+1)$	$2n(n+1)$

Hence, for both parities of $n \geqslant 4$ we obtain $s_P = 2n(n+1) = L$. For $n = 1, 2, 3$ the sum over the tables of Appendices A–C gives easily the same result. □

Next, we connect the values of $\Delta_{s_i+L}(F)$ and $\Delta_{s_i}(F)$.

Lemma 4.3. *Put* $\Pi := \prod_{l=0}^{P-1} v_l$. *We have:* $\Delta_{s_i+L}(F) = (-\Pi)^{s_i}\, \Delta_L(F)\, \Delta_{s_i}(F)$ *for all* $i \geqslant 0$.

Proof. By definition (4.2), $\Delta_{s_{i+P}} = (-1)^{\epsilon_{i+P}}\, v_0^{s_{i+P}} v_1^{s_{i+P}-s_1} \cdots v_{i+P-1}^{s_{i+P}-s_{i+P-1}}$. We split this expression into two parts, by setting

$$\delta := (-1)^{\epsilon_i+\epsilon_P}\, v_0^{s_{i+P}} v_1^{s_{i+P}-s_1} \cdots v_{P-1}^{s_{i+P}-s_{P-1}}, \qquad \delta' := v_P^{s_{i+P}-s_P} \cdots v_{i+P-1}^{s_{i+P}-s_{i+P-1}}.$$

Since $s_{i+P} - s_l = s_i + (s_P - s_l)$ and $\epsilon_{i+P} = \epsilon_P + \epsilon_i$ by (4.9), one has

$$\delta = (-1)^{\epsilon_i+\epsilon_P}\, \Pi^{s_i} \prod_{l=0}^{P-1} v_l^{s_P-s_l}.$$

On the other hand, using (4.8) and again (4.9), one gets

$$\delta' = \prod_{l=0}^{i-1} v_{l+P}^{s_{i+P}-s_{l+P}} = (-v_0)^{s_i} \prod_{l=1}^{i-1} v_l^{s_i-s_l} = (-1)^{s_i} \prod_{l=0}^{i-1} v_l^{s_i-s_l}.$$

Gathering all together gives

$$\begin{aligned} \Delta_{s_{i+P}} &= (-1)^{\epsilon_P}\, \Pi^{s_i} \prod_{l=0}^{P-1} v_l^{s_P-s_l} \times (-1)^{\epsilon_i}\, (-1)^{s_i} \prod_{l=0}^{i-1} v_l^{s_i-s_l} \\ &= (-\Pi)^{s_i}\, \Delta_{s_P}(F)\, \Delta_{s_i}(F), \end{aligned}$$

which, by Lemma 4.2, is the expected result. □

Our last step consists in computing the constants in Lemma 4.3.

Lemma 4.4. *For every* $n \geqslant 1$*:* (i) $\Pi = -1$*;* (ii) $\Delta_L(F) = (-1)^n$.

Proof. (i) Since $e_{l+P} = e_l$ by (4.9), we have $T_{l+P} = T_{P-1} + T_l$ for $l \geqslant 0$, hence

$$T_{l+mP} = mT_{P-1} + T_l$$

for any $l, m \geqslant 0$. But T is bounded by Lemma 4.1(i), so that $T_{P-1} = 0$, i.e. $|\Pi| = 2^{T_{P-1}} = 1$. On the other hand, as $v_l = -a_{l+1}$, the sign of Π is

$$(-1)^P \prod_{l=1}^{P} \operatorname{sgn} a_l = (-1)^{P-\#\{1\leqslant l\leqslant P:\ a_l<0\}} = (-1)^{\#\{1\leqslant l\leqslant P:\ a_l>0\}}.$$

For $n \geqslant 4$ we count the positive a_l, $l = 1, \ldots, P$, in the tables: a full block of either pass has 6 of them, and the remaining segments give

	$l \leqslant 6$	rows 7–9	last block	bridge c/d	pass 2 start	b1–b7	total
n even	3	2	1	4	2	5	17
n odd	3	2	3	1	5	5	19

which is odd in both cases. Thus $\operatorname{sgn}(\Pi) = -1$.
For $n = 1, 2, 3$ the sign is read off directly from Appendices A–C.

(ii) By (4.7), $E_j = \sum_{0\leqslant l<j} T_l$. Since $T_{l+P} = T_l$ by the proof of (i), one gets

$$E_{j+P} = \sum_{l=0}^{P-1} T_l + \sum_{l=P}^{P+j-1} T_l = E_P + E_j$$

for all j, hence $E_{j+mP} = mE_P + E_j$ for all $m \geqslant 0$. Then, as above, the fact that E is bounded forces $E_P = 0$. By (4.3), it follows that

$$\Delta_L(F) = \Delta_{s_P}(F) = (-1)^{\epsilon_P} \prod_{l=0}^{P-1} (\operatorname{sgn} v_l)^{\,s_P - s_l}$$

and eventually, since $s_P = L$ is even:

$$\Delta_L(F) = (-1)^{\theta(n)} \quad \text{with} \quad \theta(n) := \epsilon_P + \sum_{\substack{0\leqslant l\leqslant P-1 \\ v_l<0}} s_l.$$

For $n \geqslant 4$, write $n = 2t$ or $n = 2t + 1$. Along one period every k_m is an affine function of t and of the block number j (read off the values of K), so every $s_l = \sum_{m=1}^{l}(k_m + 1)$ is a polynomial in t and j, and $\theta(n)$ is obtained by exact summation over the segments above. The companion program `certify_sign.sage` carries this out symbolically and finds

$$\theta(2t) = \tfrac{160}{3}t^3 - 24t^2 + \tfrac{2}{3}t, \qquad \theta(2t+1) = \tfrac{160}{3}t^3 + 40t^2 + \tfrac{2}{3}t - 1.$$

In both cases $(\theta(n) - n)/2$ is a polynomial in t of degree 3 taking integer values at $t = 0, 1, 2, 3$, hence at every integer (Newton's forward-difference formula); so $\theta(n) \equiv n \pmod 2$ and (ii) is proved when $n \geqslant 4$.
For $n = 1, 2, 3$, (ii) is a simple finite computation (`verify_base.sage`). □

We can now conclude the proof of the (anti-)periodicity statement in Theorem B. By Lemma 4.3 and Lemma 4.4,

$$\Delta_{s_i+L}(F) = (-1)^n \Delta_{s_i}(F)$$

for all $i \geqslant 0$. Since $s_{i+P} = s_i + L$ by Lemma 4.2, an index $j \geqslant 0$ is of the form s_i if and only if $j + L$ is, and the other indices give 0 on both sides. Finally, we have proved that

$$\Delta^{(n+2)}_{j+2n(n+1)} = (-1)^n \Delta^{(n+2)}_j$$

for all $j \geqslant 0$, the aimed result.

Remark 4.5. In a forthcoming paper we show that $\Delta^{(n+2)}$ is a signed difference of two shifts of the ordinary Hankel sequence: for all $n \geqslant 1$ and $j \geqslant 0$,

$$\Delta^{(n+2)}_j = \epsilon_n (-1)^{nj} \Big(\Delta^{(0)}_{j+n(n+1)+1} + (-1)^{n+1} \Delta^{(0)}_{j+(n+1)(n+2)} \Big), \qquad \epsilon_n = \begin{cases} 1, & 4 \mid n, \\ -1, & 4 \nmid n. \end{cases}$$

Combined with the $2n(n+1)$-(anti)periodicity of $\Delta^{(0)}$ [HP25], this gives an alternative proof of the periodicity part of Theorem B.

Appendix A. Base case $n = 1$

As for all cases, the initial triple is given by Proposition 2.7. The sequence has pre-period 1 and period 2 (from $m = 2$), which means that $m = 3$ maps back to $m = 2$; the three transitions are identities between concrete polynomials in q, computed in `verify_base.sage`. Note that all $a_m = \pm 1$ and that $\sum_m \big(k_m + 1\big) = 4 = 2n(n+1)$ over one period.

For all $m \geqslant 1$ one has $B_m = 2q^3 - q^2 + q + 1$; the other data are listed below:

$m = 1$:

$$A_m = Q, \qquad C_m = q^4, \qquad K_m = 1, \qquad D_m = 2\,q + 1$$

$m = 2$:

$$A_m = -q^2, \qquad C_m = -Qq^2, \qquad K_m = -q^2, \qquad D_m = B_m$$

$m = 3$:

$$A_m = -Q, \qquad C_m = -q^4, \qquad K_m = -1, \qquad D_m = 2\,q + 1$$

Appendix B. Base case $n = 2$

This appendix lists the 10 tuples $\big(A_m, B_m, C_m; K_m, D_m\big)$ for $m = 1, \ldots, 10$, computed by `verify_base.sage`. The period is 9 (from $m = 2$), and $m = 10$ maps back to $m = 2$.

$m = 1$:

$$A_m = -1, \qquad B_m = -q^3 + 2\,q^2 + 1, \qquad C_m = q^5, \qquad K_m = -1, \qquad D_m = 1$$

$m = 2$:

$$A_m = q^3 - q + 2, \qquad B_m = q^3 - 2\,q^2 + 1, \qquad C_m = -q^2,$$
$$K_m = 2, \qquad D_m = \frac{1}{2}\,q + 1$$

$m = 3$:

$$A_m = -\frac{1}{8}\,q^3 - \frac{3}{8}\,q + \frac{1}{4}, \qquad B_m = \frac{1}{2}\,q^4 + \frac{3}{2}\,q^2 + 1, \qquad C_m = -\frac{1}{2}\,q^5 + \frac{1}{2}\,q^3 - q^2$$
$$K_m = \frac{1}{4}, \qquad D_m = \frac{3}{2}\,q + 1$$

$m = 4$:

$$A_m = 2\,q^3 + 2\,q^2 + 6\,q + 4, \qquad B_m = -2\,q^4 - q^3 - 6\,q^2 + 1, \qquad C_m = \frac{1}{2}\,q^5 + \frac{3}{2}\,q^3 - q^2$$
$$K_m = 4, \qquad D_m = -\frac{3}{2}\,q + 1$$

$m = 5$:

$$A_m = -\frac{1}{8}\,q^3 - \frac{1}{8}\,q^2 - \frac{3}{8}\,q - \frac{1}{4}, \qquad B_m = \frac{1}{2}\,q^4 + \frac{1}{2}\,q^3 + \frac{5}{2}\,q^2 + 1$$
$$C_m = -\frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 - \frac{3}{2}\,q^3 - q^2, \qquad K_m = -\frac{1}{4}, \qquad D_m = -\frac{3}{2}\,q + 1$$

$m = 6$:

$$A_m = -2\,q^3 - 6\,q + 4, \qquad B_m = -2\,q^4 - q^3 - 6\,q^2 + 1, \qquad C_m = -\frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 - \frac{3}{2}\,q^3 - q^2$$
$$K_m = 4, \qquad D_m = \frac{3}{2}\,q + 1$$

$m = 7$:

$$A_m = \frac{1}{8}\,q^3 - \frac{1}{8}\,q + \frac{1}{4}, \qquad B_m = \frac{1}{2}\,q^4 + \frac{3}{2}\,q^2 + 1, \qquad C_m = \frac{1}{2}\,q^5 + \frac{3}{2}\,q^3 - q^2, \qquad K_m = \frac{1}{4}$$
$$D_m = \frac{1}{2}\,q + 1$$

$m = 8$:

$$A_m = 2, \qquad B_m = q^3 - 2\,q^2 + 1, \qquad C_m = -\frac{1}{2}\,q^5 + \frac{1}{2}\,q^3 - q^2, \qquad K_m = 2, \qquad D_m = 1$$

$m = 9$:

$$A_m = q^3, \qquad B_m = -q^3 + 2\,q^2 + 1, \qquad C_m = -q^2, \qquad K_m = q^3$$
$$D_m = -q^3 + 2\,q^2 + 1$$

$m = 10$:

$$A_m = 1, \qquad B_m = -q^3 + 2\,q^2 + 1, \qquad C_m = -q^5, \qquad K_m = 1, \qquad D_m = 1$$

Appendix C. Base case $n = 3$

This appendix lists the 15 tuples for $m = 1, \ldots, 15$, computed by `verify_base.sage`. The period is 14 (from $m = 2$), and $m = 15$ maps back to $m = 2$.

$m = 1$:

$$A_m = -q, \qquad B_m = -q^4 + 2\,q^3 + q^2 + 1, \qquad C_m = q^6, \qquad K_m = -q, \qquad D_m = q^2 + 1$$

$m = 2$:

$$A_m = q^4 - q^3 + 2\,q^2 - q + 2, \qquad B_m = q^4 - 2\,q^3 + q^2 + 1, \qquad C_m = -q^3, \qquad K_m = 2$$
$$D_m = \frac{1}{2}\,q + 1$$

$m = 3$:

$$A_m = -\frac{1}{8}\,q^4 + \frac{1}{8}\,q^3 - \frac{1}{4}\,q^2 + \frac{1}{8}\,q + \frac{1}{4}, \qquad B_m = \frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 + 2\,q^3 + \frac{1}{2}\,q^2 + 1$$
$$C_m = -\frac{1}{2}\,q^6 + \frac{1}{2}\,q^5 - q^4 + \frac{1}{2}\,q^3 - q^2, \qquad K_m = \frac{1}{4}, \qquad D_m = -\frac{1}{2}\,q + 1$$

$m = 4$:

$$A_m = 2, \qquad B_m = -q^4 - 3\,q^2 + 1, \qquad C_m = \frac{1}{2}\,q^6 - \frac{1}{2}\,q^5 + q^4 - \frac{1}{2}\,q^3 - q^2, \qquad K_m = 2$$
$$D_m = 1$$

$m = 5$:

$$A_m = -q^4 + q^3 - 3\,q^2 + q - 1, \qquad B_m = q^4 + 3\,q^2 + 1, \qquad C_m = -q^2, \qquad K_m = -1$$
$$D_m = q + 1$$

$m = 6$:

$$A_m = -q^4 - q^2 - q, \qquad B_m = 2\,q^5 - q^4 + 4\,q^3 + q^2 + 1$$
$$C_m = -q^6 + q^5 - 3\,q^4 + q^3 - q^2, \qquad K_m = -q, \qquad D_m = 2\,q^2 - q + 1$$

$m = 7$:

$$A_m = -q^4 + q^3 - q^2, \qquad B_m = 2\,q^5 - q^4 + 2\,q^3 + q^2 + 1, \qquad C_m = -q^6 - q^4 - q^3$$
$$K_m = -q^2, \qquad D_m = 2\,q^3 + q^2 + q + 1$$

$m = 8$:

$$A_m = -q^4 - q^2 - q, \qquad B_m = 2\,q^5 - q^4 + 2\,q^3 + q^2 + 1, \qquad C_m = -q^6 + q^5 - q^4$$
$$K_m = -q, \qquad D_m = 2\,q^2 - q + 1$$

$m = 9$:

$$A_m = -q^4 + q^3 - 3\,q^2 + q - 1, \qquad B_m = 2\,q^5 - q^4 + 4\,q^3 + q^2 + 1, \qquad C_m = -q^6 - q^4 - q^3$$
$$K_m = -1, \qquad D_m = q + 1$$

$m = 10$:

$$A_m = -1, \qquad B_m = q^4 + 3\,q^2 + 1, \qquad C_m = -q^6 + q^5 - 3\,q^4 + q^3 - q^2, \qquad K_m = -1$$
$$D_m = 1$$

$m = 11$:

$$A_m = -q^4 + q^3 - 2\,q^2 + q + 2, \qquad B_m = -q^4 - 3\,q^2 + 1, \qquad C_m = -q^2, \qquad K_m = 2$$
$$D_m = -\frac{1}{2}\,q + 1$$

$m = 12$:

$$A_m = \frac{1}{8}\,q^4 - \frac{1}{8}\,q^3 + \frac{1}{4}\,q^2 - \frac{1}{8}\,q + \frac{1}{4}, \qquad B_m = \frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 + 2\,q^3 + \frac{1}{2}\,q^2 + 1$$
$$C_m = \frac{1}{2}\,q^6 - \frac{1}{2}\,q^5 + q^4 - \frac{1}{2}\,q^3 - q^2, \qquad K_m = \frac{1}{4}, \qquad D_m = \frac{1}{2}\,q + 1$$

$m = 13$:

$$A_m = 2\,q, \qquad B_m = q^4 - 2\,q^3 + q^2 + 1, \qquad C_m = -\frac{1}{2}\,q^6 + \frac{1}{2}\,q^5 - q^4 + \frac{1}{2}\,q^3 - q^2$$
$$K_m = 2\,q, \qquad D_m = q^2 + 1$$

$m = 14$:

$$A_m = q^4, \qquad B_m = -q^4 + 2\,q^3 + q^2 + 1, \qquad C_m = -q^3, \qquad K_m = q^4$$
$$D_m = -q^4 + 2\,q^3 + q^2 + 1$$

$m = 15$:

$$A_m = q, \qquad B_m = -q^4 + 2\,q^3 + q^2 + 1, \qquad C_m = -q^6, \qquad K_m = q, \qquad D_m = q^2 + 1$$

Appendix D. Pass 1

This appendix lists the tuples of pass 1, indexed by $m = 9j + r$ ($r \in \{1, \dots, 9\}$), starting from the initial tuple y_1 of (3.7) at $m = 1$. The formulas hold for every $n \geqslant 4$, even or odd. Each generic row is written with the range on which its triple (A, B, C) satisfies (3.4), i.e. is polynomial with $B(0) = 1$, $C(0) = 0$ (see Remark 3.4 for the exact ranges).[2] A row $\to$ the next one exactly where both triples satisfy (3.4) (see §3.1), which determines where pass 1 ends (Remark 3.4). The rows $j = 0$ are used for every $n \geqslant 4$.

$m = 1$, $m = 9j + 1$, $j = 0$:

$$A_m = -q^{n-2}, \qquad B_m = -\frac{\big(q^2 - 3\,q + 1\big)q^n + Q}{q - 1}, \qquad C_m = q^{n+3}, \qquad K_m = -q^{n-2}$$
$$D_m = -\frac{Q - q^n}{q - 1}$$

$m = 9j + 1$, $j \geqslant 1$, $n - 2j \geqslant 0$:

$$A_m = -\frac{Q(q-2)^2 q^{4j} + Q(q-2)q^{2j}\,(1 + q^n) + q^{n+1}}{(q-1)^2 q^{2j}}, \qquad B_m = -\frac{Q\,\big(2(q-2)q^{2j} + q^n + 1\big)}{q - 1}$$
$$C_m = -Qq^{2\,j}, \qquad K_m = 2, \qquad D_m = -\frac{1}{2}\,q + 1$$

$m = 2$, $m = 9j + 2$, $j = 0$:

[2] This is stronger than polynomiality: rows 4, 8, 9 of pass 1 and rows 3, 8, 9 of pass 2 are still polynomial at $n - 2j = h_r - 1$, but there $C(0) \neq 0$ (rows 4, 8 of pass 1 have $C = -1$, row 9 has $C(0) = 1$; rows 3, 8 of pass 2 have $C = -Q$, row 9 has $C(0) = -1$).

$$A_m = \frac{\left(q^2-2q+2\right)q^n + Q(q-2)}{q-1}, \qquad B_m = \frac{\left(q^2-3\,q+3\right)q^n - Q}{q-1}, \qquad C_m = -q^n$$
$$K_m = 2, \qquad D_m = \frac{1}{2}\,q+1$$

$m = 9j+2$, $j \geqslant 1$, $n-2j \geqslant 1$:

$$A_m = \frac{Qq(q-2)q^{2j}\,(1+q^n) + Qq^3q^{4j} + (q-2)^2q^n}{8q(q-1)^2q^{2j}}$$
$$B_m = \frac{Qq^2(q-2)q^{4j} + Q\left(q^2-2q+2\right)q^{2j}\,(1+q^n) + q(q-2)q^n}{2(q-1)^2q^{2j}}$$
$$C_m = \frac{Qq^2(q-2)^2q^{4j} + Qq^2(q-2)q^{2j}\,(1+q^n) + q^3q^n}{2(q-1)^2q^{2j}}, \qquad K_m = -\frac{1}{4}, \qquad D_m = -\frac{1}{2}\,q+1$$

$m = 3$, $m = 9j+3$, $j = 0$:

$$A_m = -\frac{Qq(q+2) + (q-2)\left(q^2+2\right)q^n}{8q(q-1)}, \qquad B_m = \frac{Q\left(q^2-2\right) + \left(q^3-2q^2+4q-2\right)q^n}{2(q-1)}$$
$$C_m = -\frac{q^2\left(Q(q-2) + \left(q^2-2q+2\right)q^n\right)}{2(q-1)}, \qquad K_m = \frac{1}{4}, \qquad D_m = -\frac{1}{2}\,q+1$$

$m = 9j+3$, $j \geqslant 1$, $n-2j \geqslant 3$:

$$A_m = 2\,q^{-2\,j+n-3}, \qquad B_m = -\frac{Qq\,q^{2j}\,(1+q^n) + 2(q-2)q^n}{q(q-1)q^{2j}}$$
$$C_m = \frac{Qq^2(q-2)q^{2j}\,(1+q^n) + Qq^4q^{4j} + q(q-2)^2q^n}{2(q-1)^2q^{2j}}, \qquad K_m = 2\,q^{-2\,j+n-3}$$
$$D_m = -\frac{Qq^{2\,j+1} - q^n}{(q^2-q)q^{2\,j}}$$

$m = 4$, $m = 9j+4$, $j = 0$:

$$A_m = 2\,q^{n-3}, \qquad B_m = -\frac{\left(q^3-q^2+3\,q-4\right)q^n + Qq}{q^2-q}$$
$$C_m = \frac{q\left(Qq(q+2) + (q-2)\left(q^2+2\right)q^n\right)}{2(q-1)}, \qquad K_m = 2\,q^{n-3}, \qquad D_m = -\frac{Qq-q^n}{q^2-q}$$

$m = 9j+4$, $j \geqslant 1$, $n-2j \geqslant 2$:

$$A_m = \frac{Qq^2(q-1)q^{4j} + Qq\,q^{2j}\,(1-q^n) - (q-1)q^n}{q(q-1)q^{2j}}, \qquad B_m = -\frac{Qq\,q^{2j}\,(1-q^n) - 2(q-1)q^n}{q(q-1)q^{2j}}$$
$$C_m = -q^{-2\,j+n-1}, \qquad K_m = -1, \qquad D_m = 1$$

$m = 5$, $m = 9j+5$, $j = 0$:

$$A_m = \frac{Q^2 q - \left(q^3 - q^2 + 2\,q - 1\right) q^n}{q^2 - q}, \qquad B_m = \frac{\left(q^3 - q^2 + 3\,q - 2\right) q^n - Qq}{q^2 - q}$$
$$C_m = -q^{n-1}, \qquad K_m = -1, \qquad D_m = q + 1$$

$m = 9j + 5$, $j \geqslant 1$, $n - 2j \geqslant 0$:

$$A_m = Qq^{2\,j-1}, \qquad B_m = -\frac{Q\left(2q(q-1)q^{2j} - q^n + 1\right)}{q - 1}$$
$$C_m = \frac{Qq^3(q-1)q^{4j} + Qq^2 q^{2j}\,(1 - q^n) - q(q-1)q^n}{(q-1)q^{2j}}, \qquad K_m = q^{2\,j-1}, \qquad D_m = \frac{q^{2\,j+1} - 1}{q - 1}$$

$m = 6$, $m = 9j + 6$, $j = 0$:

$$A_m = \frac{Qq(q+1) - \left(q^2 + 1\right) q^n}{q - 1}, \qquad B_m = -\frac{Q\left(2q^3 + 1\right) - \left(2q^3 - q^2 + 3q - 1\right) q^n}{q - 1}$$
$$C_m = \frac{q\left(Q^2 q - \left(q^3 - q^2 + 2q - 1\right) q^n\right)}{q - 1}, \qquad K_m = -q, \qquad D_m = Q$$

$m = 9j + 6$, $j \geqslant 1$, $n - 2j \geqslant 0$:

$$A_m = -\frac{Qq^3 q^{4j} - Qq\,q^{2j}\,(1 + q^n) + q^n}{(q-1)^2 q^{2j}}, \qquad B_m = \frac{Q\left(2q^2 q^{2j} - q^n - 1\right)}{q - 1}, \qquad C_m = -Qq^{2\,j+1}$$
$$K_m = q, \qquad D_m = -q + 1$$

$m = 9j + 7$, $j \geqslant 0$, $n - 2j \geqslant 4$:

$$A_m = q^{-2\,j+n-4}, \qquad B_m = -\frac{Qq\,q^{2j}\,(1 + q^n) - 2q^n}{q(q-1)q^{2j}}$$
$$C_m = \frac{Qq^5 q^{4j} - Qq^3 q^{2j}\,(1 + q^n) + q^2 q^n}{(q-1)^2 q^{2j}}, \qquad K_m = q^{-2\,j+n-4}, \qquad D_m = -\frac{Qq^{2\,j+2} - q^n}{(q^3 - q^2)q^{2\,j}}$$

$m = 9j + 8$, $j \geqslant 0$, $n - 2j \geqslant 3$:

$$A_m = \frac{Qq^4(q-1)q^{4j} - Qq^2 q^{2j}\,(1 - q^n) - (q-1)q^n}{q^2(q-1)q^{2j}}$$
$$B_m = -\frac{Qq^2 q^{2j}\,(1 - q^n) + 2(q-1)q^n}{q^2(q-1)q^{2j}}, \qquad C_m = -q^{-2\,j+n-2}, \qquad K_m = 1, \qquad D_m = 1$$

$m = 9j + 9$, $j \geqslant 0$, $n - 2j \geqslant 1$:

$$A_m = -Qq^{2\,j}, \qquad B_m = \frac{Q\left(2q^2(q-1)q^{2j} + q^n - 1\right)}{q - 1}$$
$$C_m = -\frac{Qq^4(q-1)q^{4j} - Qq^2 q^{2j}\,(1 - q^n) - (q-1)q^n}{(q-1)q^{2j}}, \qquad K_m = -q^{2\,j}$$
$$D_m = \frac{q^{2\,j+2} - 1}{q - 1}$$

Appendix E. Pass 2

This appendix lists the tuples of pass 2, indexed by $m = 9j + r$ ($r \in \{1, \dots, 9\}$) and written in terms of q^{2j} and q^n. Here m is an index local to pass 2: where pass 2 sits in the sequence, and at which j its rows are read, is described in §3.3. The formulas hold for every $n \geqslant 4$, even or odd. As in Appendix D, each row is written with the range on which its triple (A, B, C) satisfies (3.4). A bound $j \geqslant \frac{1}{2}$ means $j \geqslant 1$ when the rows are read at integer j (even n).

$m = 9j + 1$, $j \geqslant 1$, $n - 2j \geqslant 0$:

$$A_m = \frac{Qq\left(q^{2j} + (q^{2j} - 1)q^n\right) - q^{4j+2}}{(q-1)^2 q^{2j}}, \qquad B_m = -\frac{Q(q^n + 1) - 2q^{2j+1}}{q - 1}, \qquad C_m = -q^{2\,j}$$
$$K_m = q, \qquad D_m = -q + 1$$

$m = 9j + 2$, $j \geqslant \frac{1}{2}$, $n - 2j \geqslant 3$:

$$A_m = Qq^{-2\,j+n-3}, \qquad B_m = -\frac{Q\left(q^{2j} + (q^{2j} - 2)q^n\right)}{(q-1)q^{2j}}$$
$$C_m = -\frac{Qq^3\left(q^{2j} + (q^{2j} - 1)q^n\right) - q^{4j+4}}{(q-1)^2 q^{2j}}, \qquad K_m = q^{-2\,j+n-3}, \qquad D_m = -\frac{q^{2\,j+1} - q^n}{(q^2 - q)q^{2\,j}}$$

$m = 9j + 3$, $j \geqslant \frac{1}{2}$, $n - 2j \geqslant 2$:

$$A_m = \frac{q^2(q-1)q^{4j} + Q\left((q^{2j+1} - q + 1)q^n - q^{2j+1}\right)}{(q^2 - q)q^{2j}}$$
$$B_m = -\frac{Q\left(q^{2j+1} - (q^{2j+1} - 2q + 2)q^n\right)}{(q^2 - q)q^{2j}}, \qquad C_m = -Qq^{-2\,j+n-1}, \qquad K_m = 1$$
$$D_m = 1$$

$m = 9j + 4$, $j \geqslant \frac{1}{2}$, $n - 2j \geqslant 0$:

$$A_m = -q^{2\,j-1}, \qquad B_m = \frac{2q(q-1)q^{2j} + Q(q^n - 1)}{q - 1}$$
$$C_m = -\frac{(q^4 - q^3)q^{4j} - Q\left(q^{2j+2} + (q^2 - q - q^{2j+2})q^n\right)}{(q-1)q^{2j}}, \qquad K_m = -q^{2\,j-1}$$
$$D_m = -\frac{Q - q^{2\,j+1}}{q - 1}$$

$m = 9j + 5$, $j \geqslant \frac{1}{2}$, $n - 2j \geqslant 0$:

$$A_m = -\frac{q(q-2)^2 q^{4j} + Q(q-2)q^{2j}(1 + q^n) + Qq^n}{(q-1)^2 q^{2j}}, \qquad B_m = -\frac{2q(q-2)q^{2j} + Q(q^n + 1)}{q - 1}$$
$$C_m = -q^{2\,j+1}, \qquad K_m = 2, \qquad D_m = -\frac{1}{2}\,q + 1$$

$m = 9j + 6$, $j \geqslant 0$, $n - 2j \geqslant 2$:

$$A_m = \frac{Q(q-2)\left(q^{2j+2}+(q^{2j+2}+q-2)q^n\right)+q^{4j+5}}{8q^2(q-1)^2q^{2j}}$$
$$B_m = \frac{(q^4-2q^3)q^{4j}+Q(q^2-2q+2)q^{2j}(1+q^n)+Q(q-2)q^n}{2(q-1)^2q^{2j}}$$
$$C_m = \frac{(q^5-4q^4+4q^3)q^{4j}+Qq^2(q-2)q^{2j}(1+q^n)+Qq^2q^n}{2(q-1)^2q^{2j}}, \qquad K_m = -\frac{1}{4}$$
$$D_m = -\frac{1}{2}\,q+1$$

$m = 9j+7$, $j \geqslant 0$, $n-2j \geqslant 4$:

$$A_m = 2\,Qq^{-2\,j+n-4}, \qquad B_m = -\frac{Q\left(q^{2j+2}+(q^{2j+2}+2q-4)q^n\right)}{(q^3-q^2)q^{2j}}$$
$$C_m = \frac{Q(q-2)\left(q^{2j+2}+(q^{2j+2}+q-2)q^n\right)+q^{4j+5}}{2(q-1)^2q^{2j}}, \qquad K_m = 2\,q^{-2\,j+n-4}$$
$$D_m = -\frac{q^{2\,j+2}-q^n}{(q^3-q^2)q^{2\,j}}$$

$m = 9j+8$, $j \geqslant 0$, $n-2j \geqslant 3$:

$$A_m = \frac{(q^5-q^4)q^{4j}+Q\left(q^{2j+2}-(q^{2j+2}+q-1)q^n\right)}{(q^3-q^2)q^{2j}}$$
$$B_m = -\frac{Q\left(q^{2j+2}-(q^{2j+2}+2q-2)q^n\right)}{(q^3-q^2)q^{2j}}, \qquad C_m = -Qq^{-2\,j+n-2}, \qquad K_m = -1$$
$$D_m = 1$$

$m = 9j+9$, $j \geqslant 0$, $n-2j \geqslant 1$:

$$A_m = q^{2\,j}, \qquad B_m = -\frac{2q^2(q-1)q^{2j}-Q(q^n-1)}{q-1}$$
$$C_m = \frac{(q^5-q^4)q^{4j}+Q\left(q^{2j+2}-(q^{2j+2}+q-1)q^n\right)}{(q-1)q^{2j}}, \qquad K_m = q^{2\,j}, \qquad D_m = -\frac{Q-q^{2\,j+2}}{q-1}$$

Appendix F. Bridge 1

This appendix lists the eight tuples $\mathrm{b}1,\ldots,\mathrm{b}8$ of bridge 1. They do not involve j, and the formulas hold for every $n \geqslant 4$, even or odd.

b1:

$$A_{\mathrm{b}1} = -\frac{Qq(q+1)-(q^2+1)q^n}{q-1}, \qquad B_{\mathrm{b}1} = -\frac{\left(q^3-q^2+q-2\right)q^n+Qq}{q^2-q}, \qquad C_{\mathrm{b}1} = -q^{n-2}$$
$$K_{\mathrm{b}1} = q, \qquad D_{\mathrm{b}1} = Q$$

b2:

$$A_{\mathrm{b}2}=\frac{Q^2q-\left(q^3-q^2+2\,q-1\right)q^n}{q^2-q},\qquad B_{\mathrm{b}2}=-\frac{Q(2q^3+1)-(2\,q^3-q^2+3\,q-1)q^n}{q-1}$$
$$C_{\mathrm{b}2}=\frac{Qq^3(q+1)-q^2(q^2+1)q^n}{q-1},\qquad K_{\mathrm{b}2}=-1,\qquad D_{\mathrm{b}2}=q+1$$

b3:

$$A_{\mathrm{b}3}=-q^{n-3},\qquad B_{\mathrm{b}3}=\frac{\left(q^3-q^2+3\,q-2\right)q^n-Qq}{q^2-q}$$
$$C_{\mathrm{b}3}=\frac{Q^2q^2-\left(q^4-q^3+2\,q^2-q\right)q^n}{q-1},\qquad K_{\mathrm{b}3}=-q^{n-3},\qquad D_{\mathrm{b}3}=-\frac{Qq-q^n}{q^2-q}$$

b4:

$$A_{\mathrm{b}4}=-\frac{Qq(q+2)+(q-2)(q^2+2)q^n}{q^2-q},\qquad B_{\mathrm{b}4}=-\frac{\left(q^3-q^2+3\,q-4\right)q^n+Qq}{q^2-q}$$
$$C_{\mathrm{b}4}=-q^{n-1},\qquad K_{\mathrm{b}4}=2,\qquad D_{\mathrm{b}4}=-\frac{1}{2}\,q+1$$

b5:

$$A_{\mathrm{b}5}=\frac{(q^2-2\,q+2)q^n+Q(q-2)}{8\,(q-1)},\qquad B_{\mathrm{b}5}=\frac{Q(q^2-2)+(q^3-2\,q^2+4\,q-2)q^n}{2\,(q-1)}$$
$$C_{\mathrm{b}5}=\frac{Qq^2(q+2)+q(q-2)(q^2+2)q^n}{2\,(q-1)},\qquad K_{\mathrm{b}5}=\frac{1}{4},\qquad D_{\mathrm{b}5}=\frac{1}{2}\,q+1$$

b6:

$$A_{\mathrm{b}6}=2\,q^{n-2},\qquad B_{\mathrm{b}6}=\frac{\left(q^2-3\,q+3\right)q^n-Q}{q-1},\qquad C_{\mathrm{b}6}=-\frac{q^2\left(Q(q-2)+(q^2-2q+2)q^n\right)}{2\,(q-1)}$$
$$K_{\mathrm{b}6}=2\,q^{n-2},\qquad D_{\mathrm{b}6}=-\frac{Q-q^n}{q-1}$$

b7:

$$A_{\mathrm{b}7}=q^{n+1},\qquad B_{\mathrm{b}7}=-\frac{\left(q^2-3\,q+1\right)q^n+Q}{q-1},\qquad C_{\mathrm{b}7}=-q^n,\qquad K_{\mathrm{b}7}=q^{n+1}$$
$$D_{\mathrm{b}7}=-\frac{\left(q^2-3\,q+1\right)q^n+Q}{q-1}$$

b8:

$$A_{\mathrm{b}8}=q^{n-2},\qquad B_{\mathrm{b}8}=-\frac{\left(q^2-3\,q+1\right)q^n+Q}{q-1},\qquad C_{\mathrm{b}8}=-q^{n+3},\qquad K_{\mathrm{b}8}=q^{n-2}$$
$$D_{\mathrm{b}8}=-\frac{Q-q^n}{q-1}$$

Appendix G. Bridge 2

This appendix lists the five tuples c1, ..., c5 of bridge 2. They do not involve j, and the formulas hold for every $n \geqslant 4$, even or odd.

c1:

$$A_{\mathrm{c1}} = \frac{Qq^n + (2q-1)(q-2)}{8\,(q-1)}, \qquad B_{\mathrm{c1}} = \frac{2\,q^3 + Qq^{n+1} - 3\,q^2 + 2\,q - 2}{2\,(q-1)}$$
$$C_{\mathrm{c1}} = \frac{q^2(2q^2 - q + 2) + Q(q-2)(q+2)q^n}{2\,(q-1)}, \qquad K_{\mathrm{c1}} = -\frac{1}{4}, \qquad D_{\mathrm{c1}} = \frac{3}{2}\,q + 1$$

c2:

$$A_{\mathrm{c2}} = \frac{2\left(2q(q^2 - q - 1) + Q(q+1)q^n\right)}{q(q-1)}, \qquad B_{\mathrm{c2}} = -\frac{4q^3 - 7q^2 - q + 1 + Q(2q+1)q^n}{q-1}$$
$$C_{\mathrm{c2}} = \frac{q^2\left((2q-1)(q-2) + Qq^n\right)}{2\,(q-1)}, \qquad K_{\mathrm{c2}} = 4, \qquad D_{\mathrm{c2}} = -2\,q + 1$$

c3:

$$A_{\mathrm{c3}} = \frac{1}{2}\,Qq^{n-3}, \qquad B_{\mathrm{c3}} = \frac{Qq^n - Qq}{q^2 - q}, \qquad C_{\mathrm{c3}} = -\frac{2q^2(q^2 - q - 1) + Qq(q+1)q^n}{2\,(q-1)}$$
$$K_{\mathrm{c3}} = \frac{1}{2}\,q^{n-3}, \qquad D_{\mathrm{c3}} = \frac{q^n - q}{q^2 - q}$$

c4:

$$A_{\mathrm{c4}} = \frac{2q(q^2 - q - 1) + Q(q+1)q^n}{4\,(q^2 - q)}, \qquad B_{\mathrm{c4}} = \frac{Qq^n - Qq}{q^2 - q}, \qquad C_{\mathrm{c4}} = -Qq^{n-1}$$
$$K_{\mathrm{c4}} = \frac{1}{2}, \qquad D_{\mathrm{c4}} = -2\,q + 1$$

c5:

$$A_{\mathrm{c5}} = -\frac{2\,(Qq^n + (2q-1)(q-2))}{q-1}, \qquad B_{\mathrm{c5}} = -\frac{4q^3 - 7q^2 - q + 1 + Q(2q+1)q^n}{q-1}$$
$$C_{\mathrm{c5}} = -\frac{2q^2(q^2 - q - 1) + Qq(q+1)q^n}{2\,(q-1)}, \qquad K_{\mathrm{c5}} = 4, \qquad D_{\mathrm{c5}} = \frac{3}{2}\,q + 1$$

Appendix H. Bridge 3

This appendix lists the three tuples d1, d2, d3 of bridge 3. They do not involve j, and the formulas hold for every $n \geqslant 4$, even or odd.

d1:

$$A_{\mathrm{d1}} = \frac{Qq^n - q}{q-1}, \qquad B_{\mathrm{d1}} = -\frac{Q\left((q-2)q^n + q\right)}{q^2 - q}, \qquad C_{\mathrm{d1}} = -Qq^{n-2}, \qquad K_{\mathrm{d1}} = q$$
$$D_{\mathrm{d1}} = Q$$

d2:

$$A_{\mathrm{d2}} = -Qq^{n-1}, \qquad B_{\mathrm{d2}} = \frac{Q\big((2q-1)q^n-1\big)}{q-1}, \qquad C_{\mathrm{d2}} = \frac{q^3-Qq^{n+2}}{q-1}, \qquad K_{\mathrm{d2}} = -q^{n-1}$$
$$D_{\mathrm{d2}} = \frac{(2\,q-1)q^n-1}{q-1}$$

d3:

$$A_{\mathrm{d3}} = -\frac{Qq^n-q}{q-1}, \qquad B_{\mathrm{d3}} = \frac{Q\big((2q-1)q^n-1\big)}{q-1}, \qquad C_{\mathrm{d3}} = -Qq^{n+1}, \qquad K_{\mathrm{d3}} = -q$$
$$D_{\mathrm{d3}} = Q$$

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